

Vortex-induced vibration of separated box girders for long-span cable-supported bridges: A review

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ABSTRACT

Separated box girders (SBGs) are frequently employed in super-long-span cable-supported bridges due to their excellent flutter performance. However, SBGs are susceptible to vortex-induced vibration (VIV) caused by complex flow interference between the upstream and downstream deck. Prolonged VIV harms pedestrian comfort, traffic safety, and fatigue life of bridges. This paper aims to provide a comprehensive overview of previous studies on the VIV of SBGs, focusing on research methods, influencing parameters, excitation mechanisms and aerodynamic control measures. Scholars have conducted extensive analyses of aerodynamic performances, vortex-induced response characteristics, and flow patterns of various bridge decks and proposed a range of aerodynamic control measures to suppress the VIV of SBGs. However, the complex wind environment of super-long-span bridges and the intense aerodynamic interference between the box girders present ongoing challenges in understanding and mitigating the VIV. A thorough investigation of the various influence parameters on VIV will significantly enhance academic comprehension and practical application of SBGs. In particular, the impact of the Reynolds number on the VIV needs further clarification, and there is a lack of systematic study on the effect of aerodynamic shapes on VIV. Understanding the VIV excitation mechanisms of the SBGs is crucial for optimizing the aerodynamic shapes of bridge decks and proposing effective aerodynamic control measures. Therefore, the VIV mechanism, particularly for triple-box girders, requires additional clarification, with a specific focus on the influence of parameters such as Reynolds number and wind turbulence. As traditional passive control measures struggle to meet the demand for VIV control when the main spans of the bridge exceed 2000 m, there is an urgent need to explore novel and efficient aerodynamic control measures. Machine learning (ML) techniques hold promise in establishing vortex-induced force models, VIV warning systems, aerodynamic shape optimization, and active control of VIV.

1. Introduction

Separated box girders (SBGs) are commonly adopted in long-span cable-supported bridges due to their excellent aerodynamic performance concerning flutter instability [88,115] (Fig. 1 and Table 1). However, SBGs are more susceptible to vortex-induced vibration (VIV) than streamlined single-box girders because of the complex aerodynamic interference between the upstream deck and the downstream one [20,33]. Despite detailed examinations conducted by bridge designers and researchers during the design stage, VIV occurs frequently in actual bridges. Vertical VIVs have occurred in three super-long-span suspension bridges, i.e., Xihoumen Bridge in China, Yi Sun-sin Bridge in South Korea, and 1915 Çanakkale Bridge in Turkey [43,60,99].

VIV is a common phenomenon of flow-induced vibration (FIV)

caused by periodic shedding vortices around the surface of a bluff body. VIV occurs while the vortex shedding frequency is close to the structural natural frequency. Super-long span cable-supported bridges have more potential VIV modes in the range of frequently encountered wind speeds, increasing the risk of VIV [60]. Although VIV does not lead to catastrophic damage to bridges similar to flutter, prolonged VIV endangers pedestrian comfort, traffic safety, and fatigue life of bridges, which tends to accelerate fatigue damage and failure [33,59]. As SBGs are predominantly chosen for the proposed bridge schemes exceeding 2000 m [31], the VIV of SBGs has emerged as a constraining factor in construction of super-long-span bridges.

Scholars have conducted a series of studies to understand the vortex-induced response (VIR) characteristics, elucidate the parameters that influence the VIV, uncover the mechanism behind the VIV, and propose control measures to suppress the VIV of SBGs [55,59,64,68,71]. Fig. 2(a)

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Nomenclature

AOA	angle of attack ($^{\circ}$).
B	chord length of the individual box (m).
BG	bare girder.
$\bar{C}_D$	mean drag force coefficient.
CFD	computational fluid dynamics.
CGPs	central grid plates.
$\bar{C}_L$	mean lift force coefficient.
$\bar{C}_M$	mean moment coefficient.
$\bar{C}_P$	mean pressure coefficient.
C_D'	fluctuating drag force coefficient.
C_L'	fluctuating lift force coefficient.
C_M'	fluctuating moment coefficient.
C_P'	fluctuating pressure coefficient.
D	depth of the bridge deck (m).
DNS	direct numerical simulation.
DOF	degrees of freedom.
B/D	aspect ratio.
f_n	natural frequency of structure (Hz).
FSI	fluid-structure interaction.
f_v	vortex shedding frequency (Hz).
FIV	flow-induced vibration.
G	width of the gap (m).
G/D	gap ratio.
GVs	guide vanes.
HR	handrail.
L	length of the main span (m).
LES	large eddy simulation.
M	mass per unit length.
ML	machine learning.
MR	maintenance rail.

m^*	mass ratio, $m^* = 4M/\rho\pi D^2$.
PIV	particle image velocimetry.
RANS	Reynolds averaged Navier-Stokes.
Re	Reynolds number, $Re = \rho U_0 D / \mu$.
SBGs	separated box girders.
Sc	Scruton number, $Sc = \pi(m^*\zeta)/2$.
SHMS	structural health monitoring system.
SVM	simplified vortex model.
St	Strouhal number, $St = f_v D / U_0$.
T	torsional.
T.I.	turbulence intensity (%).
T.L.	turbulence integral length scale.
TMD	tuned mass damper.
TMDI	tuned mass damper inerter.
UC	under construction.
U_0	freestream velocity (m/s).
U_{cr}	flutter critical wind speed.
U_r	reduced velocity, $U_r = U_0 / (f_n D)$.
V	vertical.
VCSs	vertical central stabilizers.
VIF	vortex-induced force.
VIV	vortex-induced vibration.
VIR	vortex-induced response.
VGs	vortex generators.
W	width of the bridge deck.
WB	wind barrier.
Y/D	dimensionless vertical amplitude.
α	torsional angle ($^{\circ}$).
ζ	damping ratio (%).
μ	dynamic viscosity (N·s/m ²).
ρ	fluid density (kg/m ³).

illustrates the range of selected journals and the distribution of literature in this paper. Fig. 2(b) depicts the distribution of the VIV studies in SBGs with years.

The research on VIV of SBGs commenced in the 1990s, coinciding with the design proposals for the Gibraltar Bridge and the Messina Straits Bridge [6,56]. However, the application of SBGs in long-span bridges and the prominent VIV problem of the Xihoumen Bridge attracted the attention of scholars until 2008. Despite significant progress in recent years in the research related to VIV of SBGs, further research is still required due to the complexity of the problem.

This paper aims to provide scholars with comprehensive insight and detailed references on the VIV of SBGs. Part II introduces the research methods of VIV of SBGs. Part III analyzes the effects of influencing

parameters on the VIV of SBGs. Part IV discusses the vertical and torsional VIV excitation mechanisms of SBGs. Part V summarizes the aerodynamic control measures for VIV of SBGs. Finally, Part VI presents the conclusion and prospects of this paper.

2. Research methods

As illustrated in Fig. 3, field monitoring [59,60], wind tunnel tests [50,117], theoretical analysis [94,131], and computational fluid dynamics (CFD) [3,14], machine learning (ML) [30,64] are five methods in the VIV investigations of SBGs to clarify dynamics response, influencing factors and excitation mechanism and proposal effective control measures.

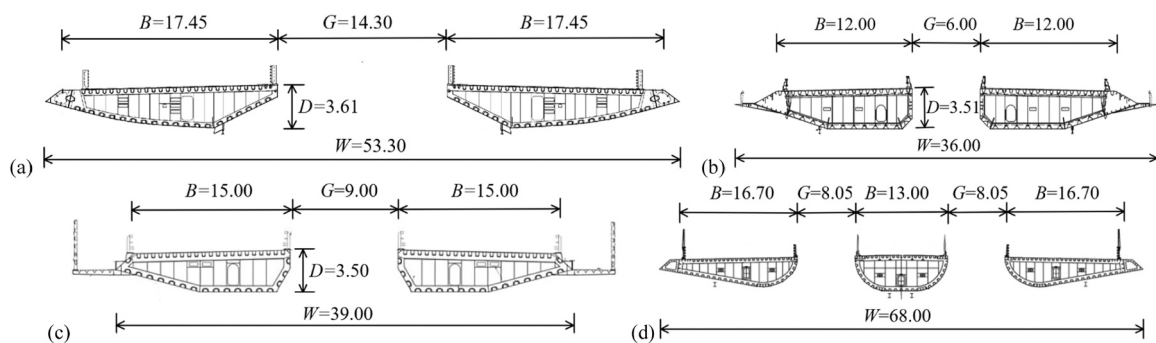


Fig. 1. Typical cross-section of bridge with SBGs: (a) Stonecutters Bridge [55], (b) Xihoumen Bridge [32], (c) 1915 Çanakkale bridge [46], (d) Xihoumen Rail-cum-Road Bridge [113].

As shown in Fig. 2(b), the wind tunnel test is the primary method used in the existing VIV studies. With the widespread installation of structural health monitoring systems (SHMS) in long-span bridges, data from field monitoring on the VIR of SGBs have been accumulated. Several researchers have proposed VIF models for SGBs that consider nonlinear characteristics. Additionally, CFD methods have also been increasingly used. Furthermore, machine learning (ML) undoubtedly provides new research perspectives and methods. Tables 2–6 summarizes some field monitoring, wind tunnel testing, and CFD studies, including static and vibration modeling. A brief overview of the application of the various research methods is provided below.

2.1. Field monitoring

Field monitoring of FIV for bridges is commonly carried out using SHMSs [61]. It is considered the most optimal and reliable method to evaluate FIV as it can capture the dynamic behaviors of real bridges in actual wind conditions [59,61]. However, field monitoring poses challenges due to its dependence on the external environment and the bridge structure, as well as its high implementation costs. The formation mechanism of VIV is intricate and influenced by numerous parameters. Although field monitoring provides accurate VIR characteristics of bridges, it cannot perform comprehensive analyses of the various influencing parameters. Additionally, gathering thorough field data is difficult due to the limited sensor arrangement.

The availability of field monitoring data has played a crucial role in advancing the understanding of VIV in SGBs. Through long-term monitoring, field pressure measurements, and visualization studies, Li et al. [59] and Li et al. [60] investigated the VIR characteristics of the Xihoumen Bridge and made initial inferences about the VIV mechanism in the full-scale bridge. Li et al. [59] found that VIV is more likely to occur in a low wind speed range of 6–10 m/s, with the wind direction nearly perpendicular to the bridge line and low turbulence intensity. The field visualization test (Fig. 4) revealed that large coherent vortex structures directly travel across the lower side of the gap without alternating shedding of Kármán-like vortices in the gap. A comparison between the 1:25 section model and field monitoring indicated that the section model test overestimated flow separation (Fig. 5) [60].

In summary, the field measurements provide the most realistic VIR, and the perfect field monitoring system helps to reveal the VIV characteristics and occurrence conditions of VIV. SHMSs combined with advanced equipment help to accurately determine the actual triggering conditions of VIV and their flow field. More field measurements provide a basis for revealing the VIV mechanism of SGBs and proposing efficient control measures.

Table 1
Summary of long-span cable-supported bridges with SGBs.

Types	Name	Year of construction	Country	Section type	L (m)	W (m)	D (m)	G/D	U_{cr} (m/s)	VIV
cable-stayed bridge (≥ 500 m)	Puyi Highway Jiajiang Bridge	2020	China	Twin-box	500	54.4	4	2.55	–	
	Yangmeizhou Bridge	UC ^a	China	Twin-box	658	–	–	–	–	
	Taoyaomen Rail-cum-Road Bridge	UC	China	Triple-box	666	65.6	4.5	1.67	–	
	Gaolan Port Bridge	UC	China	Twin-box	700	–	–	–	–	
	Huangmaohai Bridge	UC	China	Twin-box	720	–	–	–	–	
	Shanghai Yangtze River Bridge	2009	China	Twin-box	730	51	4	2.5	> 132	
	Second Wuhu Yangtze River Bridge	2017	China	Twin-box	806	53	3.5	4.86	> 100	
	Stonecutters Bridge	2009	China	Twin-box	1018	53.3	3.61	3.96	> 95	
suspension bridge (≥ 1000 m)	Xihoumen Rail-cum-Road Bridge	UC	China	Triple-box	1488	68	5	1.61	> 92.8	
	Yi Sun-sin Bridge	2012	South Korea	Twin-box	1545	29.1	2.76	1.56	> 100	✓
	Xihoumen Bridge	2009	China	Twin-box	1650	36	3.51	1.71	> 88	✓
	1915 Çanakkale Bridge	2022	Turkey	Twin-box	2023	43.9	3.5	2.29	> 64	✓

^a UC-under construction.

2.2. Wind tunnel test

Wind tunnel testing is the most commonly employed method for investigating VIV in long-span bridges. Incoming flow and model parameters can be changed easily in wind tunnel tests, and with the development of particle image velocimetry (PIV) technology, wind tunnel tests can directly obtain flow field visualization results [117,58, 68,72]. However, this testing method encounters several challenges. Firstly, due to the limitations of wind tunnel size and blockage ratio requirements, reduced-scale models must be used in experiments, which makes it difficult to meet certain similarity criteria, particularly the Reynolds similarity criterion. Secondly, traditional passive wind tunnel tests cannot simulate the turbulent characteristics under natural conditions in short test sections of wind tunnels because the turbulent has not been fully developed. Although active control wind tunnels have been developed, they are expensive and still not widely used in bridge wind engineering [33,65]. Finally, minor modifications to the aerodynamic shape and structural parameters of the bridge decks require a series of additional wind tunnel tests, raising wind tunnel test costs [122].

2.3. Computational fluid dynamics

With the advancements in algorithms and computational power, CFD has gained increasing popularity in studying the VIV of SGBs [3,14]. CFD methods offer a distinct advantage in analyzing the VIV mechanism as they can simultaneously capture the wind pressures acting on the bridge deck, aerodynamic forces, the flow field around the bridge deck, and the VIR of the bridge deck. However, the application of CFD in bridge wind engineering is still in its early stages. Europe, Japan and China have made recommendations for the use of CFD in structural wind resistance [1,11,26]. Nonetheless, a limited number of nations permit the exclusive use of CFD for structural wind resistance design [1]. Furthermore, the wide benchmark studies on the aerodynamics of a stationary 5:1 rectangular cylinder (BARC) provide the foundation for applying CFD methods to bridge aerodynamics [7]. To validate the accuracy of models and demonstrate the dependability of predicted results, experimental data from wind tunnel tests or field monitoring remains essential. Additionally, due to computational limitations, accurate turbulence simulation methods such as direct numerical simulation (DNS) and large eddy simulation (LES) are challenging to apply in practical engineering under high Reynolds numbers.

Some scholars have studied the reliability of CFD in investigating the flow characteristics of stationary SGBs [2,3,81]. Álvarez et al. [2] utilized 2D URANS simulation to comprehend the vortex-shedding

mechanism around twin-box girders. However, at higher angles of attack (AOAs) of 7° and 10° , the pressure distribution on the upper surface of the box girder deviated significantly from experimental results due to the inaccurate representation of the strong suction using the SST $k-\omega$ turbulence model. Álvarez et al. [3] further investigated the influence of the connecting crossbeam between two boxes on the flow characteristics of SBGs using 3D LES.

Recently, scholars have used two-dimensional fluid-structure interaction (FSI) simulation to reveal the driving mechanism of VIV in SBGs, which initially provides a theoretical basis for the design of aerodynamic stability and vibration control of separated box girders. Dong et al. [22] used the SST $k-\omega$ turbulence model to investigate the spanwise correlation of aerodynamic forces on stationary and vibrating SBGs, indicating that self-excited forces around the box girders increased the spanwise correlation of aerodynamic forces. Chen et al. [14] and Duan et al. [23] used the SST $k-\omega$ turbulence model and found that the interaction of the vortices shed from the upper and lower surfaces on the gap side was considered to be the main cause of VIV. Duan et al. [23] found that the flow restrictor plates at the bottom of the gap side effectively suppressed the vertical VIV, which significantly reduced the airflow interactions at the gap side. Based on the experimental data of Wang et al. [98], Zhang et al. [121] used 2D RANS simulations to further elucidate the differences in flow characteristics and aerodynamic work distribution between linear and curved webs during the vertical VIV of a separated three-box girder.

In summary, the majority of CFD studies currently in existence are based on stationary models. However, it is necessary to conduct FSI simulations to identify the VIV excitation mechanism of SBGs because the interaction between the bridge deck and the flow field in the lock-in region cannot be accurately described by stationary models. FSI simulation utilizing a relatively high-precision CFD method, such as LES, facilitates a detailed understanding of the VIV mechanism of SBGs.

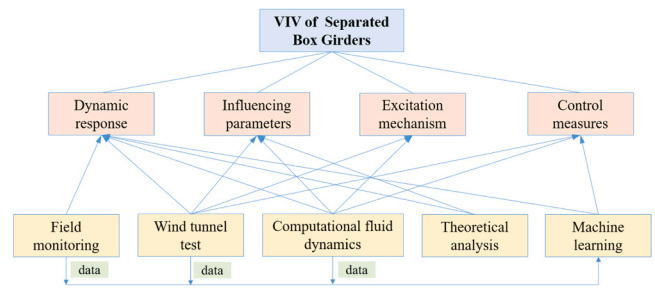


Fig. 3. The research methods in the VIV investigations of SBGs.

2.4. Theoretical analysis

Due to the complexity of bluff body flows, it is difficult to directly derive analytical expressions for VIV from the Navier-Stokes equations and structural motion equations. Currently, the prevailing method involves the development of simplified semi-empirical models through wind tunnel experiments. The crucial aspect of these models lies in accurately determining the vortex-induced force (VIF) model. Scanlan's empirical nonlinear model is the most classical VIF model in the study of bridge VIV [89]. This model attempts to reproduce the self-limiting characteristic of VIV by using a nonlinear item of aerodynamic damping force defined as a product function of the velocity and displacement square of the vibration [25]. However, Scanlan's model does not accurately describe the nonlinear VIF acting on SBGs.

Diana et al. [20] used wind tunnel tests to determine the transfer function (flutter derivatives) between VIF and the structural motion of the Mainland Bridge. They proposed a single-of-freedom nonlinear time-domain numerical model. However, the transfer function definition is linear, which is insufficient for expressing the high nonlinear forces induced by vortex shedding in the lock-in region. Zhu et al. [131]

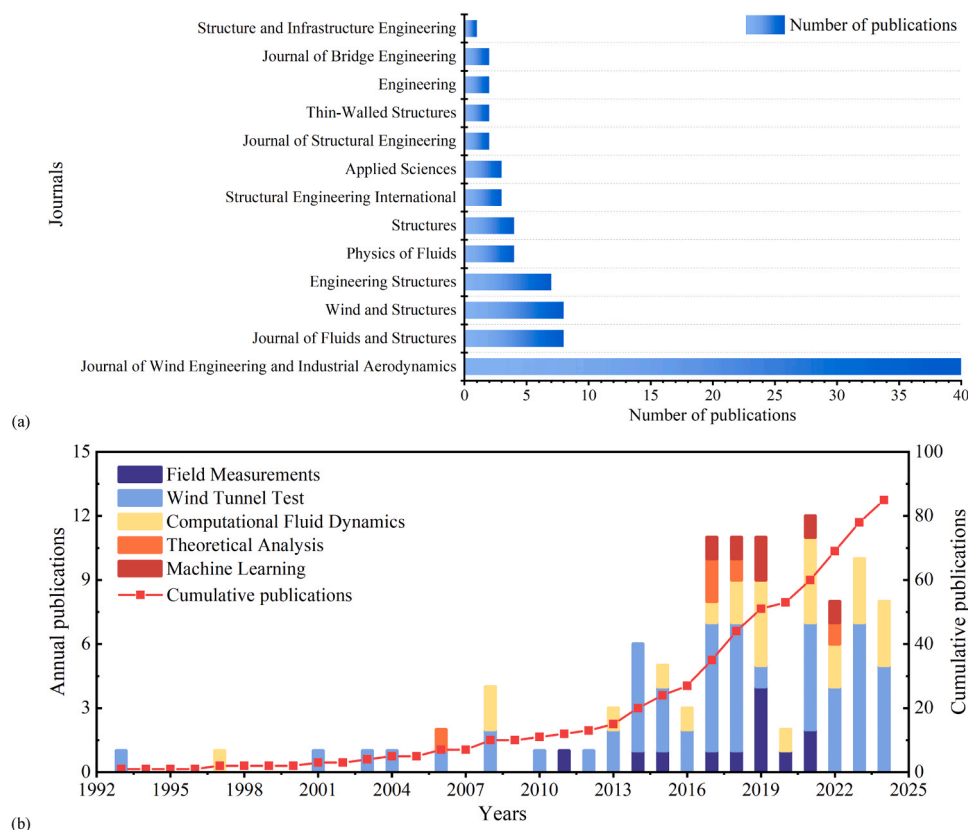


Fig. 2. Statistics related to the VIV study on SBGs. (a) Distribution of published academic journals. (b) Number of publications from 1993 to August 2024.

Table 2
Experimental studies on stationary SBGs.

Researchers	Section type	Re× 10 ⁻⁴	B/D	G/D	AOA (°)	T.I. (%)	Deck attachments				Slope	Measurements
							BG ^a	HR ^b	MR ^c	WB ^d		
Matsuda et al.[76]	Twin-box	1.1–150	–	–	–5–5	–		✓				St, $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$
Kwok et al.[47]	Twin-box	4.4	5.6	0–6	–10–10	–	✓				✓	St, $\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$
Diana et al.[19]	Triple-box	6.9–31	–	–	–10–10	–		✓		✓	✓	$\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, $\overline{C}_P$
Chen et al.[13]	Twin-box	1	4.3	0–3.4	0	–	✓				✓	$\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, Flow patterns
Lee et al.[57]	Twin-box	3.8/ 6.7–46.7	4.5	1.6	–5–5	< 0.5		✓			✓	$\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$
Kargarmoakhar et al. [44]	Twin-box	26–120	5.1	2.7	–6–6	–		✓				St, $\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$
Meng et al.[78]	Triple-rec	1–2.2	3.7	0–10.3	0	0	✓					St, $\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, Flow patterns
Meng et al.[80]	Triple-box	0.08	3.7/ 2.5	1.5	–5–5	< 0.4	✓				✓	St, Flow patterns
Meng et al.[77]	Triple-box	2.8	3.7/ 2.5	1.5	0	< 0.4	✓				✓	St, $\overline{C}_P$, C_P' , Flow patterns

^a BG-Bare girder;

^b HR-Handrail;

^c MR-Maintenance rail;

^d WB-Wind barrier.

Table 3
Numerical studies on stationary SBGs.

Researchers	Section type	Re× 10 ⁻⁴	B/D	G/D	AOA (°)	T.I. (%)	Deck attachments				Slope	Measurements
							BG	HR	MR	WB		
Watanabe and Fumoto [101]	Twin-box	1	5.8	2.5	–16–16	0		✓	✓		✓	$\overline{C}_P$, $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, Flow patterns
Miranda et al.[81]	Twin-box	1	4.3	0.9–3.4	0	0	✓				✓	$\overline{C}_P$, C_P' , Flow patterns
Laima et al.[48]	Twin-box	0.02–1000	4.3	1.7	0	0	✓				✓	St, $\overline{C}_P$, Flow patterns
Álvarez et al.[2]	Twin-box	4.7	5.4	0–9.7	–10–10	–	✓				✓	St, $\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, Flow patterns
Laima et al.[52]	Twin-box	0.02–10	4.3	1.7	0	0	✓				✓	$\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, C_D' , C_L' , C_M' , Flow patterns
Shang et al.[92]	Twin-box	4	5.6	0.3–6	–3–3	–	✓				✓	St, $\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$, $\overline{C}_M$, Flow patterns
Ran et al.[84]	Triple-box	0.02–2.8	3.7	2.5	0	0	✓				✓	St, $\overline{C}_P$, C_P' , $\overline{C}_D$, $\overline{C}_L$
Ran et al.[85]	Triple-box	0.02–2.8	3.7	2.5	0	0	✓				✓	St, $\overline{C}_P$, Flow patterns

proposed a semi-empirical model for vertical VIV of SBGs considering the influence of turbulence. Zhu et al. [132] further proposed a mode-by-mode VIV analysis method to predict the VIR of SBGs under non-uniformly distributed turbulent wind. Sun et al. [94] proposed a new parameter identification technique for distinguishing terms with the same order in the polynomial VIF model. The parameter identification results obtained from wind-tunnel tests on a twin-box deck reveal that only the Rayleigh-type damping terms are significant in the vertical VIF. However, all of the same-order nonlinear damping terms in the torsional VIF should not be overlooked. Chen et al. [15] found that the VIF of the twin-box girder can be regarded as fully correlated with a large amplitude of VIV.

Recently, CFD simulations of forced vibration combined with semi-empirical VIF models have been used to predict the VIR of bridge decks. Sarwar and Ishihara [87] proposed a prediction method for the reduced velocity corresponding to the maximum VIR based on the forced vibration. Noguchi et al. [83] further predicted the VIR in the lock-in regime. Wang and Chen [100] enhanced the predictive efficiency by proposing a continuous variable amplitude forced vibration technique. In addition, Song et al. [93] introduced a two-stage approach for VIR prediction, which involves a preliminary extraction of aerodynamic damping through rapid evanescent vibration simulations, followed by the application of the VIF model for final VIR estimation.

Additionally, utilizing the spatial distribution characteristics of the localized aerodynamic forces around the bridge deck and the associated vortex drift patterns, Hu et al. [39] and Hu et al. [38] developed a simplified vortex model (SVM) to reveal the VIV mechanism of a typical streamlined closed-box girder. Zhao et al. [125] developed a SVM for the

vertical VIV of a separated triple-box girder, but the flow pattern around it is very complex, which makes it very difficult to reveal the VIV mechanism only by the SVM.

To sum up, it is crucial to develop VIF models that consider the nonlinear characteristics of SBGs. Moreover, the development of reduced-order techniques provides possibilities for simulating the complex FSI in the VIV process of the main girder [107,110].

2.5. Machine learning

Field monitoring, wind tunnel tests, and CFD have accumulated a large amount of raw data for wind resistance of bridges, which contains rich wind engineering knowledge. However, obtaining the deep knowledge hidden in the data using simple data processing methods is difficult. Machine learning (ML), a type of data-driven method, is exactly a powerful tool for analyzing large amounts of data. ML will become an essential tool for solving challenges in future wind engineering.

Li et al. [62] utilized field monitoring data from the Xihoumen Bridge to investigate the relationship between the wind speed field and vortex-induced vibration (VIV) mode. They employed an automatic and efficient clustering algorithm to obtain this relationship quickly. Subsequently, they established a data-driven model based on the decision tree and support vector regression to directly establish the connection between wind field characteristics and VIV response [63]. Li et al. [64] proposed a deep learning framework for identifying VIR based on Scanlan’s model and VIV observed from field monitoring. However, this framework has some limitations. To simplify the network structure, they

Table 4
Experimental studies on free vibrating SBGs.

Researchers	Section type	$Re \times 10^{-4}$	B/D	G/D	AOA (°)	T.I. (%)	Deck attachments				Slope	DOFs ^a	Measurements
							BG	HR	MR	WB			
Larsen et al. [55]	Twin-box	0.3/1.3/6.6	4.8	4	0	–	✓				✓	V ^b	C_p' , Flow patterns, Y/D
Laima et al. [50]	Twin-box	1.2–4.4	4.3	1.7	0	–	✓				✓	V	St, $\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, $\bar{C}_M$, Flow patterns, Y/D
Li et al. [60]	Twin-box	3.4–5.8	4.3	1.7	–	–		✓	✓	✓	✓	V&T ^c	St, $\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, Flow patterns, Y/D , α
Liu et al. [68]	Twin-box	0.6–11.2	4.3	1.7	0	< 0.46	✓				✓	V	St, $\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, Flow patterns, Y/D
Laima and Li [49]	Twin-box	1.2–4.1	4.3	0–10.3	0	0	✓				✓	V&T	St, $\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, $\bar{C}_M$, Flow patterns, Y/D , α
Yang et al. [114]	Twin-box	0–7.7/0–154/47.3–213	4.3	1.7	–3–3	–		✓	✓	✓	✓	V&T	St, Y/D , α
Yang et al. [116]	Twin-box	1.8–3.4	4.3	0–4.3	–3–3	–	✓	✓	✓	✓	✓	V&T	Y/D , α
Yuan et al. [117]	Twin-box	0.06–0.3	4.3	1.7	0	< 0.4	✓				✓	V&T	Flow patterns, Y/D , α
Ma et al. [71]	Twin-box	0–2.3	4.7	3.7	–5–5	< 0.5		✓	✓	✓	✓	V&T	Y/D , α , Flow patterns
Laima et al. [51]	Twin-box	1.3/1.9–6.8	4.3	1.7	0	< 0.5	✓	✓	✓	✓	✓	V&T	St, $\bar{C}_p$, C_p' , Flow patterns, Y/D , α
Liu et al. [68]	Twin-box	0–3.6	4.3	1.7	–3–3	≤ 0.5		✓	✓	✓	✓	T	$\bar{C}_p$, C_p' , Flow patterns, α
Wang et al. [98]	Triple -box	0–60.8/0–147	4.2/2.9	1.7	–3–3	0.3–3	✓	✓	✓	✓	✓	V&T	Y/D , α
Yang et al. [113]	Triple -box	1.2–10.5	3.9/2.6	1.6	0	–	✓	✓	✓	✓	✓	V&T	$\bar{C}_p$, C_p' , α
Chen et al. [14]	Twin-box	0.2–1.9	4.5	2.7	–5–5	< 0.5		✓	✓	✓	✓	V	Y/D , $\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, Flow patterns
Ge et al. [30]	Twin-box	0–21.1	5.5	2.6	–3–3	0/2.8		✓		✓	✓	V&T	Y/D , α
Wang et al. [99]	Triple -box	0.4–1	4.2/2.9	1.7	–3–3	< 0.5					✓	V&T	$\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, Flow patterns, Y/D , α
Duan et al. [24]	Twin-box	0.1–2.6	4.64	1.9	–3–3	–	✓	✓			✓	V	Y/D
Meng et al. [79]	Triple -box	0.2–1.9	3.9/2.6	1.7	–5–5	< 0.4	✓	✓		✓	✓	V	St, Flow patterns, Y/D
Wu et al. [105]	Triple -box		3.9/2.6	1.6	0	–		✓	✓	✓	✓	T	$\bar{C}_p$, C_p' , Flow patterns, α
Zhao et al. [125]	Triple -box	0.0–1.4	3.9/2.6	1.7	0	< 2.0		✓	✓	✓	✓	V	$\bar{C}_p$, C_p' , Y/D
Chen et al. [15]	Twin-box	0.0–8.5	4.9	2.7	–3–3	< 1.0		✓	✓	✓	✓	V	C_p' , Y/D
Duan et al. [23]	Twin-box	0.2–1.7	4.64	1.9	–3–3	–		✓	✓		✓	V	Y/D
He et al. [36]	Twin-box	0.4–3.8	–	–	–3–3	–		✓			✓	V&T	Y/D , α

^a DOFs denotes degrees of freedom;

^b V denotes DOF of vertical;

^c T denotes DOF of torsional.

Table 5
Numerical studies on free vibrating SBGs.

Researchers	Section type	$Re \times 10^4$	B/D	G/D	AOA (°)	T.I. (%)	Deck attachments				Slope	DOFs	Measurements
							BG	HR	MR	WB			
Chen et al. [14]	Twin-box	0.6	4.5	2.7	0	0	✓				✓	V	Flow patterns, Y/D
Zhang et al. [121]	Triple -box	0.9–5.5	4.2/2.9	1.7	0	0	✓				✓	V	St, $\bar{C}_p$, C_p' , Flow patterns, Y/D
Duan et al. [23]	Twin-box	0.4–0.6	4.64	1.9	–3–3	–		✓	✓		✓	V	$\bar{C}_p$, C_p' , $\bar{C}_D$, $\bar{C}_L$, C_L' , Flow patterns, Y/D

Table 6
Field measurements on free vibrating SBGs.

Researchers	Section type	$Re \times 10^{-4}$	B/D	G/D	AOA (°)	T.I. (%)	Deck attachments				Slope	DOFs	Measurements
							BG	HR	MR	WB			
Li et al. [59]	Twin-box	145–266	4.3	1.7	–	1.8–6.6		✓	✓	✓	✓	V&T	St, $\bar{C}_p$, C_p' , Y/D
Li et al. [60]	Twin-box	145–266	4.3	1.7	–	–		✓	✓	✓	✓	V&T	St, $\bar{C}_p$, C_p' , Flow patterns, Y/D
Yang et al. [114]	Twin-box	47.3–213	4.3	1.7	–3–3	–		✓	✓	✓	✓	V&T	St, Y/D

neglected the higher-order terms of Scanlan’s model. Additionally, the framework is exclusively focused on predicting the VIR, which presents challenges in understanding the physical mechanism of FSI. Xu et al. [111] developed a VIV identification model based on a parameter

optimization algorithm to predict the occurrence of VIV. To estimate the probability of VIV for SBGs, Ge et al. [30] established a decision tree that considers the effects of structural damping ratios, wind direction, and wind speed. Additionally, Montoya et al. [82] proposed a data-driven

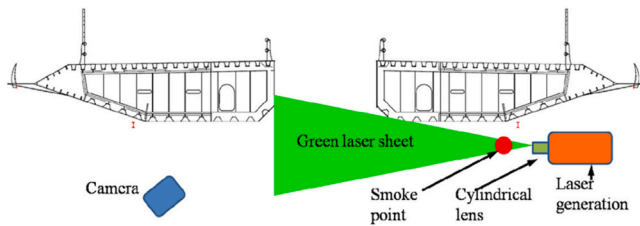


Fig. 4. Schematic of the flow visualization system used in the full-scale bridge model [60].

adaptive surrogate model optimization method based on wind tunnel test data, which can improve the VIV performance of SBGs by optimizing the gap ratio and maintenance rails position. Recently, Deng et al. [18] used deep reinforcement learning and transfer learning technology to intelligently and actively control the aerodynamic forces on the decks of long-span bridges.

To summarize, ML emerges as a highly promising tool for investigating VIV. ROM and deep learning techniques offer innovative approaches to advance the VIV theory. Moreover, utilizing deep learning and reinforcement learning frameworks helps effectively select optimal aerodynamic shapes. Additionally, reinforcement learning exhibits tremendous potential for active control of VIV phenomena.

3. Influencing parameters

As shown in Fig. 6, many factors impact the VIV performance of SBGs, such as aerodynamic shapes (e.g., aspect ratio, gap ratio, wind fairing, inclination angle of the web, corner cut), Reynolds number,

AOA, wind turbulence, mass-damping coefficients, and attachments, among others.

3.1. Aerodynamic shapes

VIV performance is susceptible to the aerodynamic shapes of bridge decks, while the gap ratio (defined as G/D) is a relatively well-studied aerodynamic shape parameter for SBGs. Studies on stationary cross-section models have shown that gap ratio significantly changes the aerodynamic performance and flow field characteristics of SBGs [13,2,47]. Kwok et al. [47] and Chen et al. [13] have found that the surface pressure on the windward side of the downstream deck turns negative to positive with the increase in the gap ratio, which leads to an increase in the drag coefficient of the bridge decks. Fig. 7 gives the variations of the St number with gap ratio for Stonecutters Bridge [2,47,92] and Xihoumen Bridge [13,48,49]. As shown in Fig. 7, when the gap ratio is less than 2.5, the variation of St for both the Xihoumen Bridge and the Stonecutters Bridge is extremely complex, suggesting that flow patterns in this regime may change drastically [49,92]. As the gap ratio continues to increase, the St number tends to be stabilized with different values

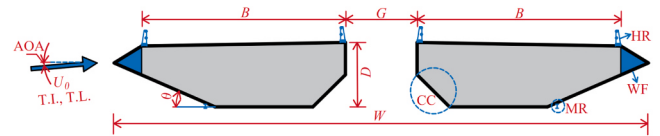


Fig. 6. Schematic diagram of influencing parameters on the VIV of SBGs (CC-corner cut; HR-handrails; MR-maintenance rail; WF-wind fairing; θ -the angle between the bottom plate and side panel).

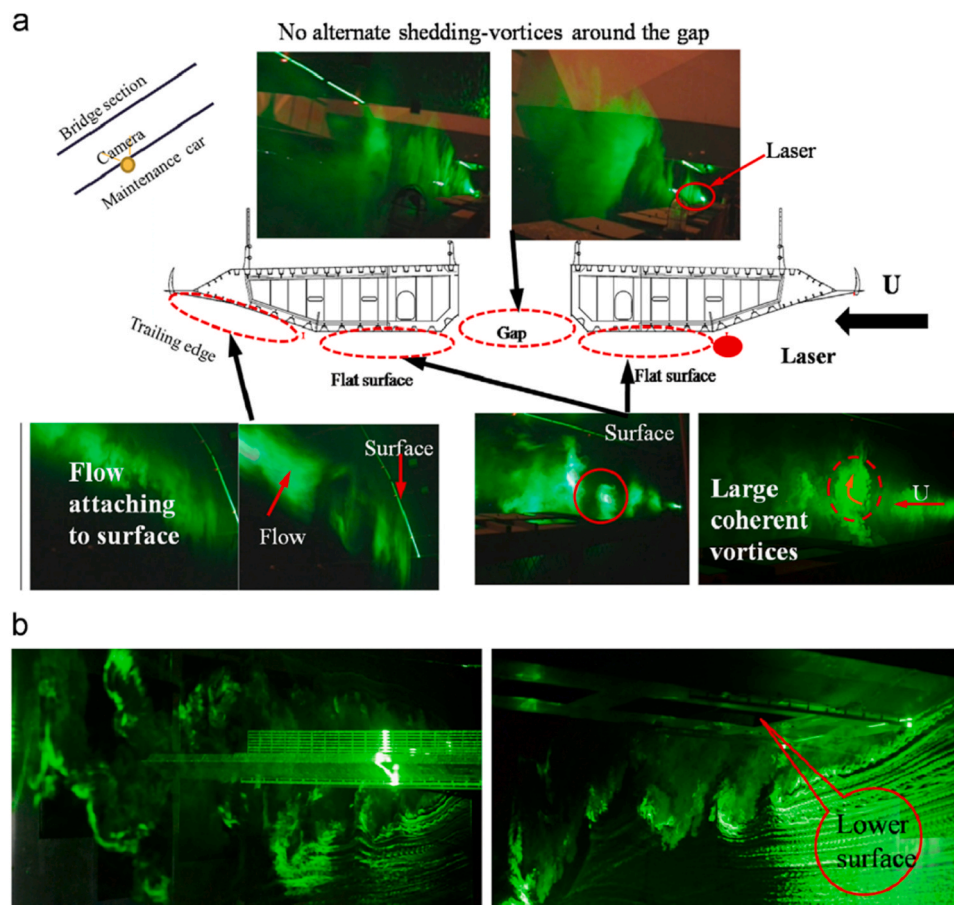


Fig. 5. Smoke visualization of the flow around the lower surface of the twin-box girder. (a) Field monitoring and (b) 1:25 section model test [60].

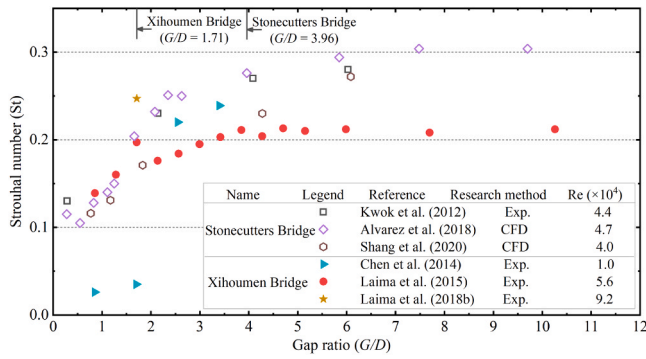


Fig. 7. Variation of St number with gap ratio for the bare girders of Stonecutters Bridge and Xihoumen Bridge [13,2,47–49,92].

between two SBGs cross-section models, which may be caused by the variations of local aerodynamic shapes or Reynolds number. Significant differences exist in the St number obtained at different Reynolds numbers even for the same bridge (Xihoumen Bridge), especially when the gap ratio is small.

Laima and Li [49] found that three flow patterns are presented around the box girders with the increasing gap ratio of the bare deck of the Xihoumen Bridge. When $vortex shedding starts to appear on the gap side at $2.138 \leq G/D \leq 10.26$, the flow of the gap controls the flow around the downstream box girders without alternate vortex shedding phenomenon occurring behind the downstream box girder. However, the Stonecutters Bridge exhibits a co-shedding regime at $2.145 \leq G/D \leq 6.028$ [92]. The variations of deck geometries may be the cause of different flow regimes. Recently, Meng et al. [78] investigated the aerodynamic performance and flow field patterns of stationary triple rectangular cylinders and categorized the flow patterns around the triple rectangular cylinders with the increase of the gap ratio into four modes.$

Owing to various flow patterns as the gap ratio grows, Laima and Li [49] found that the SBGs exhibited different types of VIV as the gap ratio varies. When $G/D < 1.710$, no VIV; $1.710 \leq G/D \leq 2.138$, vertical VIV; $2.566 \leq G/D \leq 2.933$, vertical and torsional VIV; $G/D > 3.421$, torsional VIV. The largest amplitude of vertical VIV is at $G/D = 1.710$. Furthermore, Yang et al. [116] studied the VIV performance of the Xihoumen Bridge with attachments. They observed that the vertical VIV amplitude of the main girder increased and then decreased with an increasing gap ratio. In the discussion, the existing studies only focus on the effect of the gap ratio on the VIV of the twin-box section of the Xihoumen Bridge. The flow structure analysis shows that different flow characteristics exist in various geometries of the single girder. Therefore, the vibration characteristics with the change of spacing ratio may be different for different deck geometries. The influence of spacing ratio on the VIV needs further investigation for the more deck geometries.

Apart from the gap ratio, the geometry of the single girder plays a significant role in the VIV performance of SBGs. Belloli et al. [5] examined the impact of the web inclination in the central rail box girder on the VIV performance of a separated triple-box girder. The study revealed a considerable reduction of negative surface pressure on the downstream highway box girder by increasing the web inclination, resulting in the suppression of VIV. For separated triple-box girders, Diana et al. [20] and Wang et al. [98] found that straight webs outperform curved webs with regard to VIV performance. Meng et al. [77] found that for the girder with curved webs, the upstream gap flow is dominated by the separated-shear-layer instability, and the downstream gap flow is dominated by the double-shear-layer instability, resulting in the 'dual-frequency' phenomenon. In contrast, for the girder with linear webs, the global field is dominated by the separated-shear-layer instability. Zasso et al. [118] found that the gap configuration significantly impacts VIV performance. A corner cut causes the VIV to occur at a higher velocity regime with sharply increasing

maximum amplitude and a wider lock-in region. Chen et al. [14] recently demonstrated that gap configuration has little effect on the St number but significantly impacts the maximum VIV amplitude. Additionally, recent studies have explored the effect of wind fairing shape on the flutter and static stability of SBGs [115]. Recently, our team has used LES to analyze the impact flow mechanism of the single-side fairings on the aerodynamic characteristics of the twin-box girder (Fig. 8) [21].

In conclusion, the aerodynamic shapes of bridge decks are of utmost importance in determining the flow field characteristics, aerodynamic performance, and VIV performance of SBGs. However, previous studies have only focused on specific deck geometries, which hinders the generalization of their findings due to the substantial variations in deck geometries among different SBGs. Furthermore, the Reynolds number substantially influences the impact of aerodynamic shapes on the flow field characteristics, aerodynamic performance, and VIV performance of SBGs, which has received insufficient attention in research.

3.2. Reynolds number

In the 1990s, VIVs were observed in the erection stage of the Great Belt East Bridge with wind speeds ranging between 16 and 20 m/s ($St = 0.21$), which contradicted the critical wind speed of approximately 23 m/s ($St = 0.16$) calculated from low speed (low Re) wind tunnel tests carried out during the design stage. The phenomenon attracted the attention of scholars, and they pointed out that the Reynolds number effect was the most probable cause. Schewe and Larsen [90] confirmed the significant effect of the Reynolds number on the bridge girder cross-section through a high pressure wind tunnel test. Larose and D'Auteuil [54] emphasized that the Reynolds number sensitivity of the aerodynamic performance of SBGs is related to their deck geometries, and ignoring the Reynolds number effect may lead to systematic errors. The change in separation-reattachment and wake structure with increasing Reynolds number is the essence of the Reynolds number effect on the bridge girder.

Flow pattern, aerodynamic performance and VIV of SBGs are also substantially sensitive to the change in Reynolds number [48,58]. Li

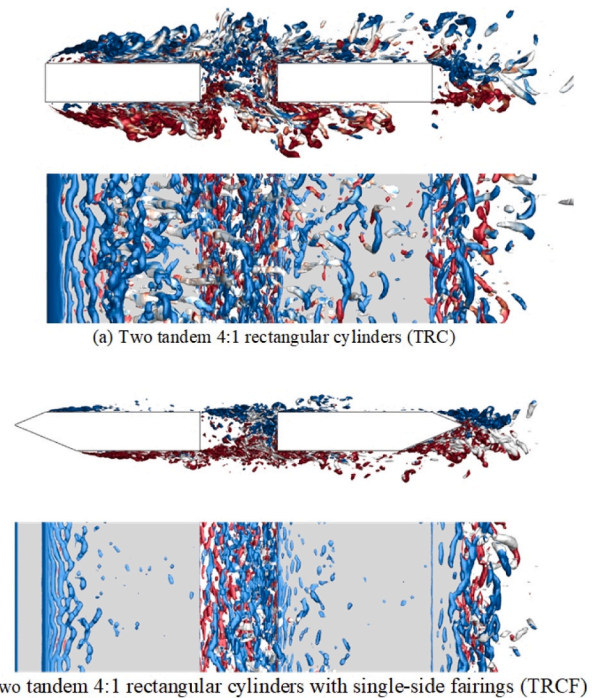


Fig. 8. Circumferential and spanwise distributions of instantaneous vortex structures around cylinders (iso-surface: Q-criterion at $Q = 4U_0^2/D^2$; red color: z-vorticity $\omega_z > 3U_0/D$; blue color: z-vorticity $\omega_z < -3U_0/D$) [21].

et al. [58] investigated the Reynolds number effect on the flow structure of the stationary section model of the Xihoumen Bridge using PIV. The results reveal that when the Reynolds number grows, the transition of the separated shear layer moves upstream, and the airflow quickly reattaches to the surface of the twin-box girder after the transition. Furthermore, Laima et al. [48] reveal that the flow patterns around the leading edge, gap, and wake change dramatically with the Reynolds number using the CFD method (Fig. 9). Recently, Ran et al. [85] found that the flow structure around the triple-box girder varies very complexly as the Reynolds number increases from 10^2 to 2.8×10^4 (Fig. 10).

Based on the study on Yi Sun-sin Bridge, Lee et al. [57] found that the Reynolds number significantly affects the surface pressure and Strouhal number of the SBGs. As shown in Fig. 11, pressure coefficients near the gap shift from negative to positive with the increase of Reynolds number. Fig. 12 summarizes the variation of the Strouhal number and drag coefficient with the Reynolds number of several SBGs [44,48,58,76], with the result of Great Belt East Bridge serving as a reference [90]. It is worth noting that both the Strouhal number and drag coefficient exhibit a substantial sensitivity to changes in the Reynolds number. The influence of the Reynolds number is more pronounced at lower Reynolds numbers (below 10^5) compared to higher Reynolds numbers (above 10^5). Furthermore, there are significant disparities in the Strouhal number and drag coefficient among bridges with different aerodynamic shapes. The effect of the Reynolds number on the Strouhal number and drag coefficient varies significantly depending on the aerodynamic shape of the bridge. Recently, Ran et al. [84] found that the drag coefficient of a triple-box girder generally tends to decrease with increasing Reynolds number. However, in the range of $Re = 5 \times 10^2 - 3 \times 10^3$, the dramatic evolution of the flow field lead to a complex change of the drag coefficient.

Although extensive studies have been conducted by Chinese scholars on the VIV of Xihoumen Bridge, the amplitude of VIV varies significantly depending on the experimental conditions [116,58,60]. Fig. 13 presents

the VIV amplitude of the Xihoumen Bridge at various Reynolds numbers from several research studies. Li et al. [60] found that the amplitudes obtained in wind tunnel tests (low Reynolds number) were considerably smaller compared to those measured in the field (high Reynolds number). In the wind tunnel tests, two types of VIV, namely vertical and torsional, were identified, whereas only vertical VIV was observed during field monitoring. Li et al. [58] further experimentally investigated the Reynolds number effect on VIV using three section models. The results indicated that higher Reynolds numbers led to lower critical reduced wind velocity, larger vibration amplitude, and wider lock-in range. However, the experimental Reynolds number range was $0.6 \times 10^4 - 11.2 \times 10^4$, 1–2 orders of magnitude smaller than the full-scale Reynolds number. In contrast, Yang et al. [116] found that the maximum amplitude at a higher Reynolds number was smaller than at a lower Reynolds number. Above all, the influence of Reynolds number on VIV is highly intricate and can be affected by various factors such as the characteristics of the incoming wind and attachments.

Additionally, Diana et al. [19] found that surface roughness significantly impacts the Reynolds number effect on the aerodynamic performance of SBGs. Lee et al. [57] found that the railings reduce the Reynolds number effect on the aerodynamic performance at negative AOA, and the boundary layer trip strip attached at the bottom surface of the deck reduces the Reynolds number dependency by fixing the flow separation.

In summary, the aerodynamic performances and VIVs of SBGs are greatly influenced by the Reynolds number, but the findings of related studies are inconsistent and even conflicting. The Reynolds number effect of VIV is notably influenced by deck geometries, and it is important to explore the impact of deck geometries and attachments on the Reynolds number effect of VIV. Moreover, the mechanisms of the Reynolds number effect on VIV need further clarification.

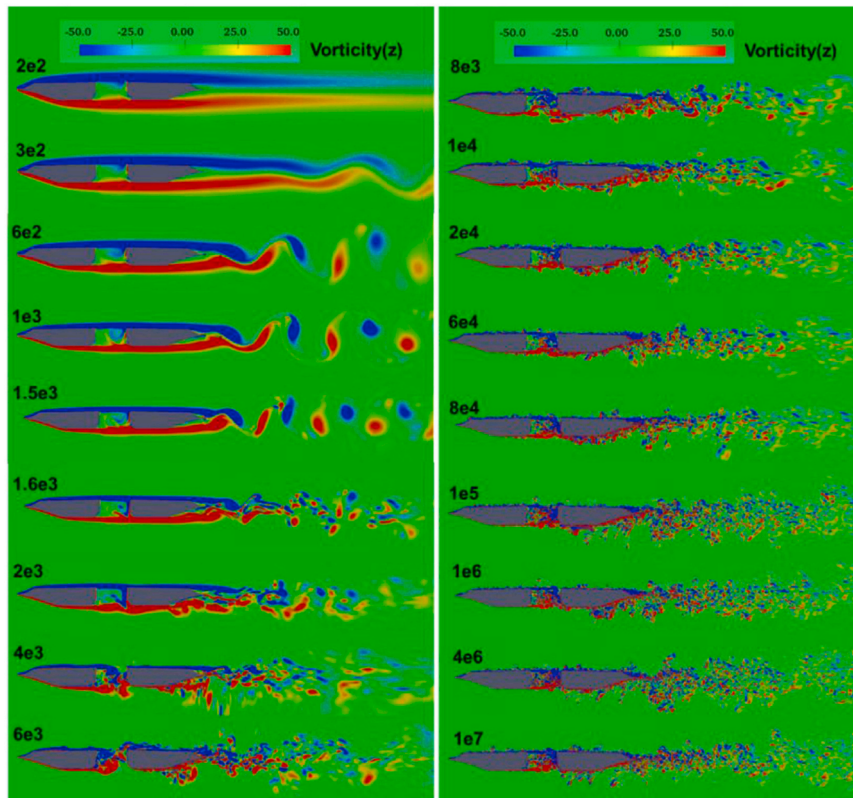


Fig. 9. Instantaneous z-direction vorticity contours at $z/D = 0.5$ section for various Reynolds numbers [48].

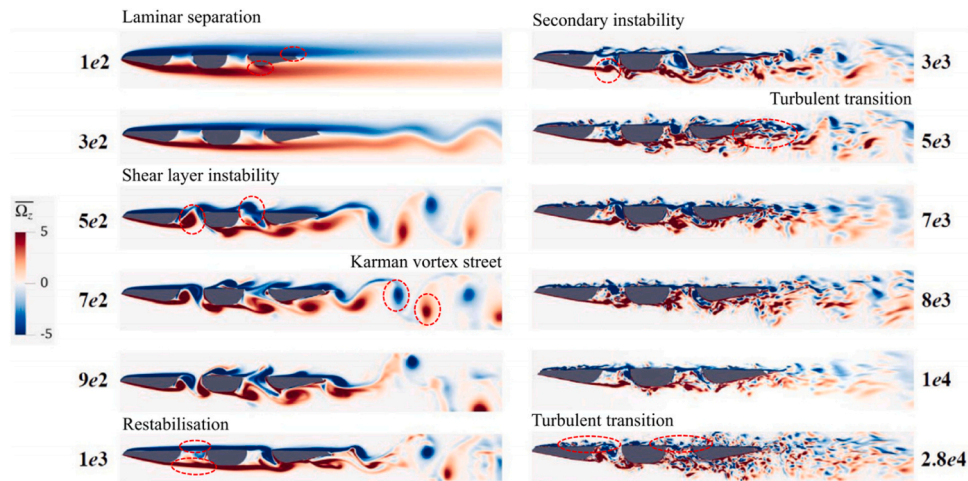


Fig. 10. Instantaneous spanwise vorticity contours in the cross-section at $z/H = 0.5$ [85].

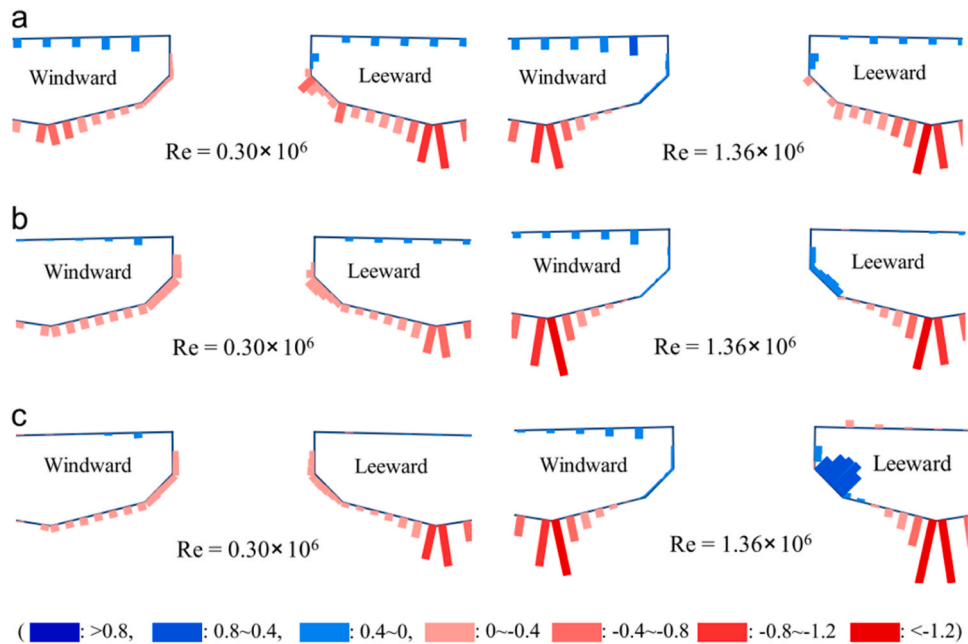


Fig. 11. Mean pressure coefficients near the gap at subcritical and supercritical Reynolds numbers: (a) -3° , (b) 0° and (c) $+3^\circ$ [57].

3.3. Angle of attack

The range of AOA at the deck location of mountainous bridges frequently exceeds $\pm 10^\circ$. Even for coastal bridges with flat topography, the daily AOA at the deck location can reach -5° to 5° [70,104]. Studies have indicated that VIVs of SBGs tend to have a most unfavorable AOA rather than AOA = 0° [30,116]. Therefore, the VIV performance of bridge decks in the range of frequently encountered AOAs should be tested at the design stage.

Studies have been conducted on the performance of VIV for SBGs at various AOAs [30,116]. Unfortunately, most existing research has focused on small AOAs (from -3° to 3°). Ma et al. [71] suggested that -3° is the most unfavorable AOA for vertical VIV, while torsional VIV only occurs at AOA = 3° . Wang et al. [99] investigated the VIV of separated triple-box girders within AOAs from -3° to 3° . Their results indicated that AOA = 0° is most unfavorable, and the driving mechanism of VIV differs at each AOA. As shown in Fig. 14, the pulsating lift of the central railroad box girder plays a significant role in VIV at AOA = 3° , whereas at AOAs = 0° and -3° , the pulsating lift of the

downstream highway box girder is the primary factor stimulating VIV. Meng et al. [79] found that the VIV onset wind speed of the separated triple-box girders increased monotonically with increasing AOA from -5° to 5° , and the maximum amplitude of VIV increased as the negative AOA grew.

Analyzing the aerodynamic performance and flow structure of stationary section models at various AOAs can assist in understanding the mechanism of VIVs of SBGs [124]. Watanabe and Fumoto [101] observed that the aerodynamic forces undergo significant changes as AOA varies, primarily due to variations in the separation-reattachment phenomenon occurring at the lower side of the leading edge of the faring. For the Stonecutters Bridge, Kwok et al. [47] found that AOA significantly affects the pressure distribution around the SBGs, and hence the corresponding aerodynamic forces of the deck, and the drag coefficient is most sensitive to the change of AOA. For the Yi Sun-sin Bridge, Lee et al. [57] observed that the negative pressure at the leading edge of the downstream box girder is significantly higher at AOA = 3° than non-positive AOAs. In addition, as illustrated in Fig. 11, the aerodynamic performance is more sensitive to the variation of Reynolds

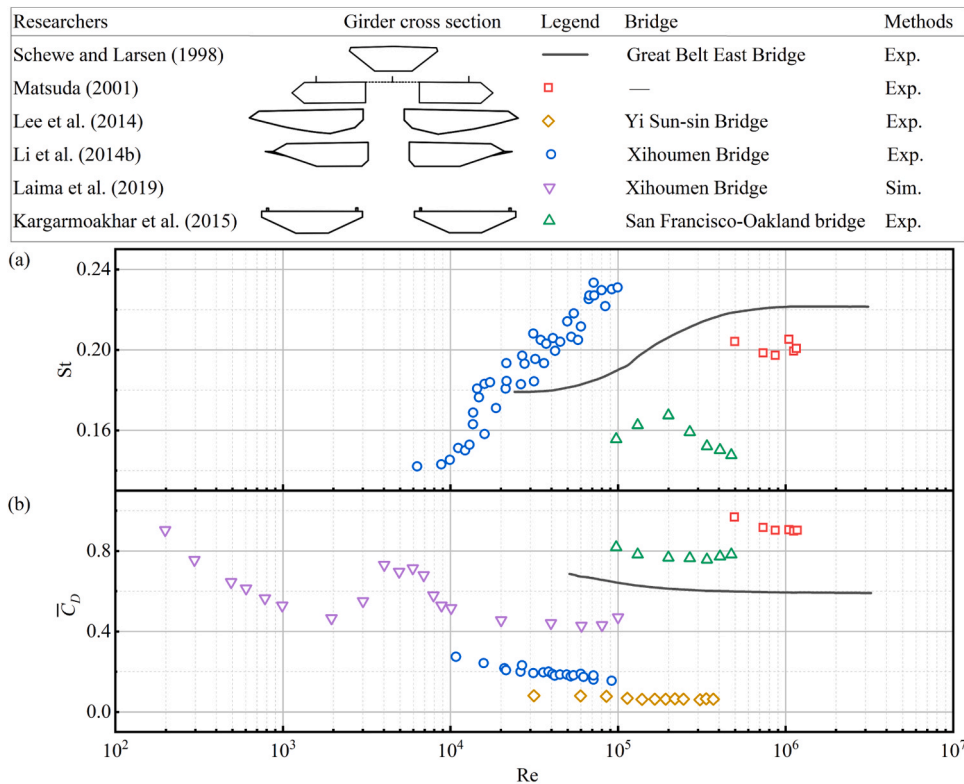


Fig. 12. Variation of (a) Strouhal number St and (b) mean drag coefficient $\bar{C}_D$ with Reynolds number [44,48,58,76,90].

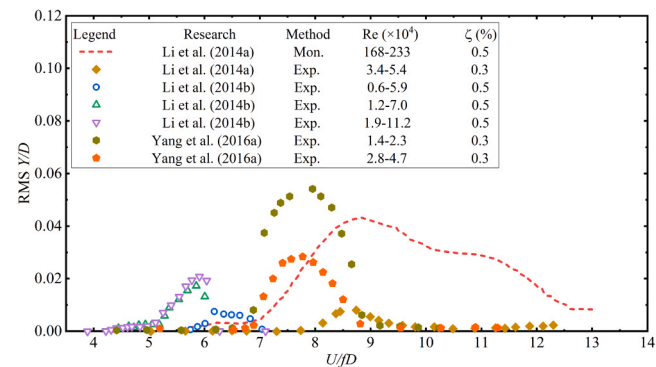


Fig. 13. The VIV amplitude of Xihoumen Bridge at various Reynolds numbers [116,58,60].

number at positive AOAs. Recently, Meng et al. [80] conducted PIV tests and found that the flow characteristics around the triple-box girder vary significantly with various AOAs. At $AOA = 0^\circ$, the upstream gap flow is dominated by separated-shear-layer instability, while the downstream gap flow is influenced by double-shear-layer instability. At $AOA = +5^\circ$, separated-shear-layer instability governs the entire flow field. At $AOA = -5^\circ$, double-shear-layer instability prevails in the flows around both gaps.

In summary, variations in aerodynamic characteristics and VIV performance occur due to changes in the aerodynamic shapes and flow interactions at the gap of SBGs under different AOA conditions. It is crucial to elucidate the VIV performance and its driving mechanism of SBGs at different AOAs, particularly in light of the nonlinear characteristics at large AOAs. Additionally, existing studies have indicated that the aerostatic and flutter stability may be worse under small yaw wind than the incident wind, which is normal for the longitudinal axis of a bridge [129]. However, the VIV performance of SBGs under yaw winds

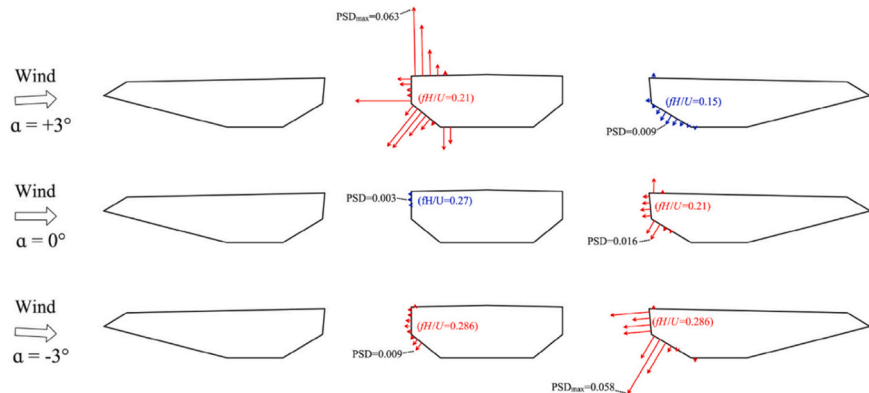


Fig. 14. PSD peak value of pressure coefficient for static bare girder [99].

has not received much attention in previous research.

3.4. Scruton number

The mass-damping parameter, also known as the Scruton number, was first introduced by Scruton for his experiments on elastically mounted circular cylinders [91,97]. A higher Scruton number generally corresponds to a lower amplitude of structural VIV and a narrower range of the lock-in region [86]. To minimize the dispersion of the Scruton number on the prediction results of the vibration response of rectangular-like structures, Vandiver [96] and Marra et al. [75] adjusted the $m^* = 4 M/(\pi \rho D B)$.

Diana et al. [19] investigated the impact of the damping ratio on the vertical VIV of the Messina Bridge. They found that the maximum amplitude of two vertical VIV regions and the range of the lock-in region decreased with the damping ratio increase but with different magnitudes of reduction. Similarly, Wu and Laima [103], in their study on the Xihoumen Bridge, observed that an increase in damping ratio leads to a decrease in VIV amplitude, a narrower range of the lock-in region, and a decrease in the folding speed corresponding to the maximum amplitude. However, Zhu et al. [130] pointed out that the damping ratio does not affect the range of the lock-in region. For the Yi Sun-sin Bridge, Kim et al. [45] found that the effect of modal mass on the amplitude is small, while the modal damping ratio is an important parameter for evaluating VIV performance. Hwang et al. [43] investigated the effect of the Scruton number on VIV modes of the bridge deck with different modal masses and damping ratios. They determined the modal damping ratios of the bridge using the frequency domain decomposition method and found that the damping ratio of the VIV-triggered mode was the lowest among the dominant modes that were considered.

Accurate estimation of modal frequencies and mechanical damping of the section model is a prerequisite for the VIV characterization studies. However, the role of aerodynamic damping is often neglected when measuring mechanical damping in current experiments, resulting in an overestimation of mechanical damping, which affects the accuracy of the results of the study [123]. Cao and Ge [8] investigated the aerodynamic damping and the added mass of a separated triple-box girder during vibration at zero wind speed. The result showed that aerodynamic damping is the main source of damping for the vibration system, and the added mass significantly affects the vibration frequency of the section model. Zhang and Xu [123] found that the added mass is insensitive to the amplitude, while the aerodynamic damping increases linearly with the amplitude and even exceeds the mechanical damping for separated twin-box girders. Therefore, the measurement of aerodynamic damping should be emphasized in the VIV study for long-span SBGs.

3.5. Wind turbulence

It is generally believed that the amplitude of VIV in the main girder tends to decrease as the wind turbulence increases [27]. There are two possible explanations for the influence of turbulence on the suppression of VIV in structures. The first explanation suggests that turbulence can enhance the dissipation of vortices, resulting in a decrease in vortex strength. The second explanation proposes that turbulence can reduce the span correlation of VIV [106].

Field monitoring conducted on Xihoumen Bridge revealed that VIV is easily triggered when the turbulence is low [40,59,60]. Additionally, the inhomogeneity of the wind field along the spanwise direction of the bridge is a significant factor affecting VIVs of the full-scale bridge [60]. Wang et al. [98] experimentally found that increasing turbulence intensity can help to reduce the VIV amplitude of a separated triple-box girder. For the bare deck of the Xihoumen Bridge, Zhu et al. [130] found that the span correlation of VIV slightly decreases with the increase of turbulence intensity, but the fundamental frequency components of VIV still exhibit good correlation at high turbulence intensity (T .

$I = 8\%$) within the lock-in region. Zhu et al. [131] found that the amplitude of vertical VIV decreases non-linearly as turbulence intensity increases. Furthermore, Zhu et al. [132] emphasized that the energy of turbulence is concentrated in the low-frequency range, suppressing better lower-order modes of VIV compared to higher-order modes of VIV. As a result, the VIV of the full-scale bridge deck often corresponds to higher-order modes. Recently, Ma et al. [73] found that increasing the turbulence intensity can effectively suppress the torsional VIV of the separated twin-box girder with large cantilever plates, but it is ineffective in suppressing vertical VIV.

In summary, increasing turbulence intensity generally helps to suppress the VIV amplitude of SBGs. Higher turbulence intensity may disrupt the coherent structures of the vortices, leading to a suppression of VIV. However, the specific mechanism through which turbulence intensity suppresses VIV in SBGs requires further clarification. For the VIV of SBGs, the turbulence integral length scale has been rarely studied due to the limitation of experimental conditions. However, studies on flow around circular cylinders have shown that the turbulence integration scale greatly influences its aerodynamic performance [119]. Considering the difference between the turbulence integral length scale of wind tunnel tests and the actual incoming wind, it is necessary to investigate the influence of the turbulence integral length scale on the VIV of SBGs. Therefore, the effect and its mechanism of turbulence on VIV necessitate exploration utilizing active control wind tunnel experiments and CFD methods.

3.6. Attachments

To ensure the safety of pedestrians and vehicles and routine maintenance, attachments such as handrails, wind barriers, and maintenance rails must be installed for the surface of the bridge deck. However, attachments seriously disturb the boundary layer of the bridge deck, affecting the aerodynamic characteristics and VIV performance.

For Xihoumen Bridge, Laima et al. [51] found that crash barriers with higher wind permeability have a minimal impact on the flow field characteristics of the bridge deck. On the other hand, windward side wind barriers and maintenance rails on the upstream box girder increase the width of separation flow and decrease the strength of the vortices around the gap, which affects the vertical VIV of the SBG (Fig. 15). Duan et al. [24] found that the vortices shedding from the windward edge and railing impinged on the top and bottom surfaces of the downstream section, resulting in the worsening of VIV performance. A numerical simulation also found that adding wind barriers causes significant changes in the aerodynamic characteristics and vortex shedding patterns of the twin-box girder [81]. Yang et al. [116] showed that wind barriers can suppress vertical VIV but increase the torsional VIV of the twin-box girder.

Recently, for a separated triple-box girder, Yang et al. [112] found that large-scale vortices shedding behind the safety barrier of the upstream highway box and the maintenance rails, as well as above the downstream highway box, producing periodic pulsating lifts and moments that result in severe vertical and torsional VIVs. Optimizing the location of maintenance rails (from the bottom of the highway box to the lower surface of the cross beam) can help improve the VIV performance. Wu et al. [105] found that wind barriers and railings on the upstream highway box girder cause strong disturbance to the flow on the highway deck surface, unsteady shear flow separating from the wind barriers top acts on the central and downstream girders, leading to a large-amplitude torsional VIV. Meng et al. [79] pointed out that attachments at the upstream box girder (highway wind barriers and maintenance rails) enhanced the VIVs, while those at the central box girder (railway wind barriers) weakened them. In addition, the results show that prioritizing the installation of highway wind barriers during the construction process can optimize the aerodynamic performance of bridges [77].

In summary, the attachments significantly change the VIV performance of the SBGs. Therefore, future studies need to optimize the

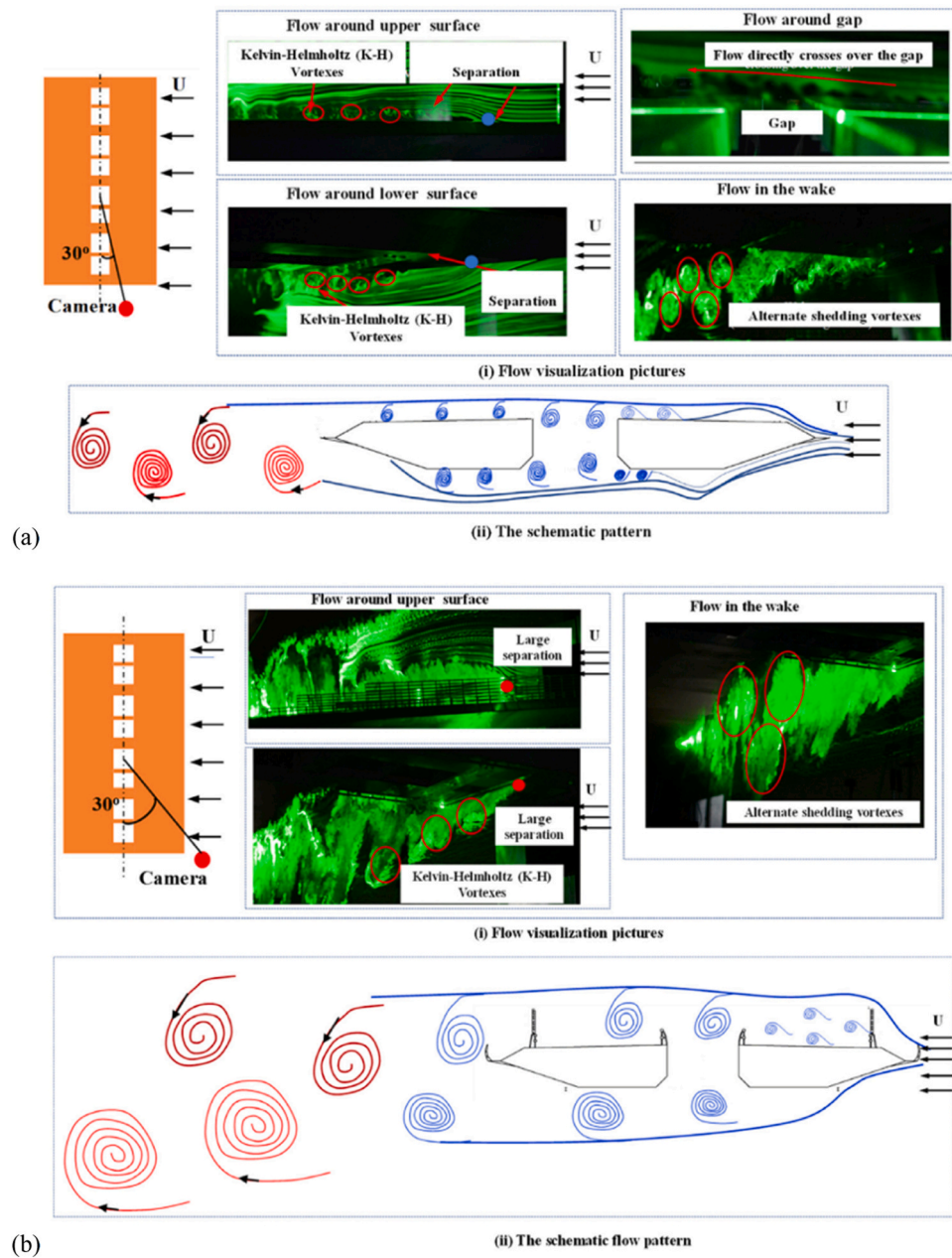


Fig. 15. Flow visualizations around twin-box girder and corresponding schematic flow patterns: (a) bare deck, (b) twin-box girder with full attachments [51].

geometric details and the placement of attachments to minimize their adverse effects on VIVs. In addition, mechanisms of interference of attachments on aerodynamic characteristics and VIV performance of SGBs and the Reynolds number effect of SGBs with attachments also require further investigation.

4. Vortex-induced vibration excitation mechanisms

Investigating the VIV excitation mechanism is crucial in order to understand the underlying causes of the phenomena, optimize the aerodynamic shapes of bridge decks, and propose effective aerodynamic control measures [29]. Scholars primarily use pressure measurement [50,68], PIV [50,117] and CFD [50,55] methods to reveal the excitation mechanism of VIV.

4.1. Vertical vortex-induced vibration excitation mechanism

It is generally agreed that the significant vertical VIV of separated twin-box girders is closely linked to the generation and shedding of large-scale vortices from the upstream box girder, which subsequently impacts the downstream box girders [117,29,71]. For Stonecutters Bridge, Larsen et al. [55] found that vortices shed from the upstream box girder will impinge on the downstream box girder, resulting in the fluctuating pressure here being 3 to 4 times larger than the upstream box girder from where the vortices are shed. They tended to render the twin-box girder more susceptible to vortex shedding excitation than the single-box girder. For Xihoumen Bridge, Laima et al. [50] found that the energy input at the onset of VIV is mainly caused by the near wake vortices shedding (2S mode, which comprises two single vortices per cycle of motion) excitation of the downstream box girder. As VIV amplitude increased, strong vortices formed in the gap. These vortices impinged on the windward of the downstream box girder and caused the

flow to separate and reattach periodically around the windward corners of the downstream box girder, which was identified as the primary energy source for the large VIV amplitude.

Furthermore, by comparing the flow characteristics of stationary and dynamic models, Yuan et al. [117] deduced that the gap vortex and model motion mutually enhance each other. They found two vertical VIV lock-in regions (V1 and V2), while the amplitude of V1 is significantly larger than that of V2. As shown in Fig. 16, in V1, the wake shedding shares a mode (both 2S modes) with gap shedding, and the vortex is shed synchronously in the wake and the gap. The gap shedding is stronger than the wake shedding. The periodic rolling up of the shear layer causes the formation of the Kármán vortex street in the gap, resulting in wake-induced vibration. They identified the central gap as an essential factor in inducing VIVs, while the wake vortex shedding of the downstream box girder is not. In V2, the vortex mainly forms at the leading corners of the downstream box girder rather than in the gap. As shown in Fig. 17, the shedding mode changes from 2S to 2(2S) mode (two pairs of vortices successively shed in one circle).

The VIV mechanism of separated triple-box girders is more complex. Researchers found that the main girder of the Messina Bridge has two vertical VIV lock-in regions. These lock-in regions have different mechanisms [5,19]. The first lock-in region is caused by the periodic shedding of vortices at a fixed separation point on the surface of the upstream box girder. The excitation of vortices on the gap between the central girder and the downstream box girder causes the second lock-in region. For the Xihoumen Rail-cum-Road Bridge, Yang et al. [112] found that large-scale vortices formed and alternately shed on the gap side, behind the maintenance rails, and above the downstream box girder. These vortices generated periodic lift forces, resulting in vertical VIV. Wang et al. [99] found that the VIV is caused by the pulsation pressure resulting from the impact of the alternately shed vortices on the gap side of the windward central box girder and the downstream box girder. The

strong motion-induced vortex on the surface of the main girder enhances the pulsation pressure, serving as the primary driving force for large amplitude VIV. Recently, Meng et al. [79] found a 'dual-frequency' phenomenon in two gaps of a static triple-box girder and pointed out that the 'dual-frequency' phenomenon may increase the lock-in region. Kármán vortex of the first gap and motion-induced vortex of the downstream highway box girder separation during vibration are essential in the onset and peak of VIV amplitudes, respectively.

4.2. Torsional vortex-induced vibration excitation mechanism

Compared to vertical VIV, the excitation mechanism of torsional VIV has received less attention, as it has only occurred in wind tunnel tests and has yet to be observed on full-scale bridges. For the bare deck of Xihoumen Bridge, Yuan et al. [117] found that both vortex excitation on the gap and Kármán vortex shedding from the downstream box girder have essential effects on two torsional VIV lock-in regions. After adding attachments, Laima et al. [51] showed that large-scale Kármán vortex shedding alternately at the end of the downstream box girder is the primary cause of torsional VIV.

In recent studies, scholars explored the distribution of distributed aerodynamic forces contributing to VIV of SBGs, as shown in Fig. 18. Liu et al. [68] found that the main cause of torsional VIV is the distributed aerodynamic forces in the rear region of the upper and lower surfaces of the downstream box girder and the front region of the upper surface of the upstream box girder. Unlike streamlined closed-box girders [37], central slotting increases the correlation of the aerodynamic forces in the upper and lower surfaces of the downstream box girder, enhancing the overall VIV. For a separated triple-box girder, Yang et al. [113] found that the distributed VIV formed on the lower surface of the upstream highway box girder, the gap between the upstream highway box girder and the central railroad box girder, the upper surface of the downstream

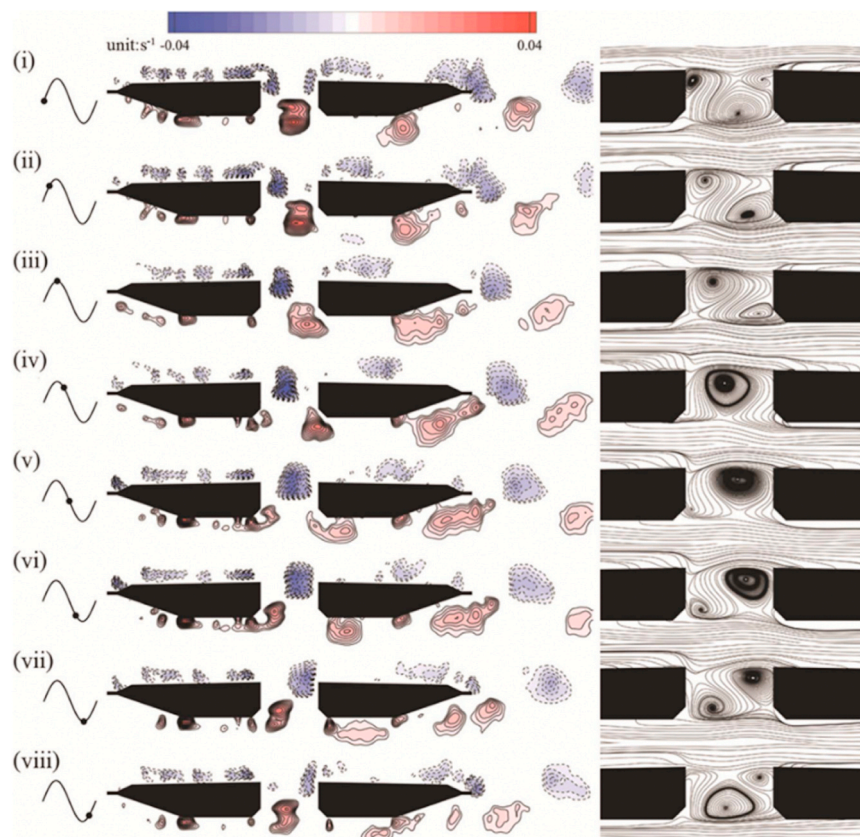


Fig. 16. Evolution of the flow fields around the dynamic model in V1 while the inflow velocity is 0.66 m/s [117].

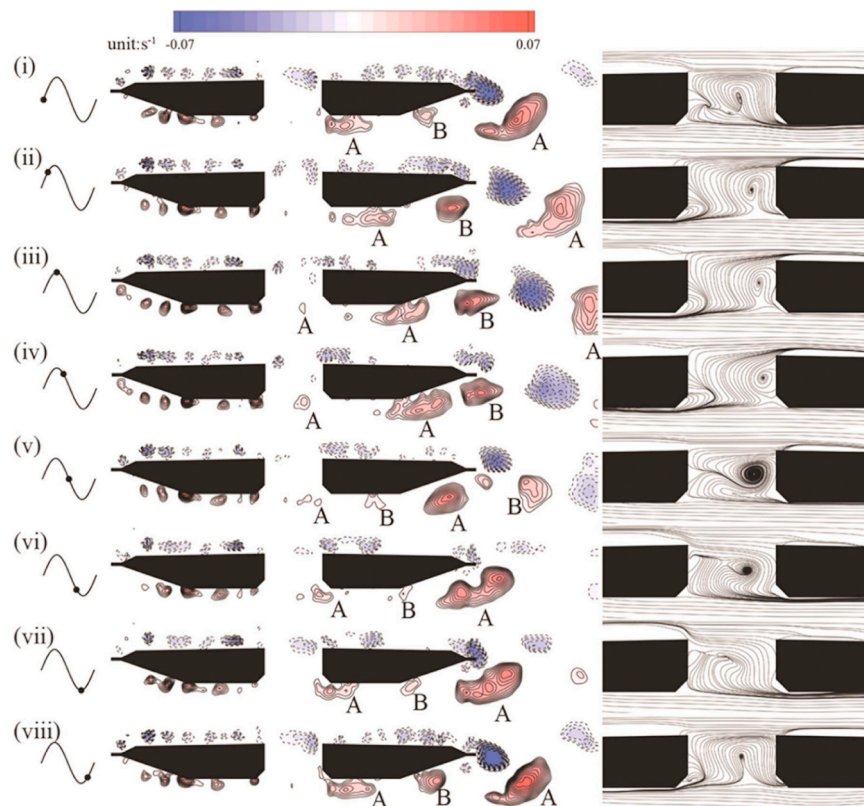


Fig. 17. Evolution of the flow fields around the dynamic model in V2 while the inflow velocity is 1.14 m/s [117].

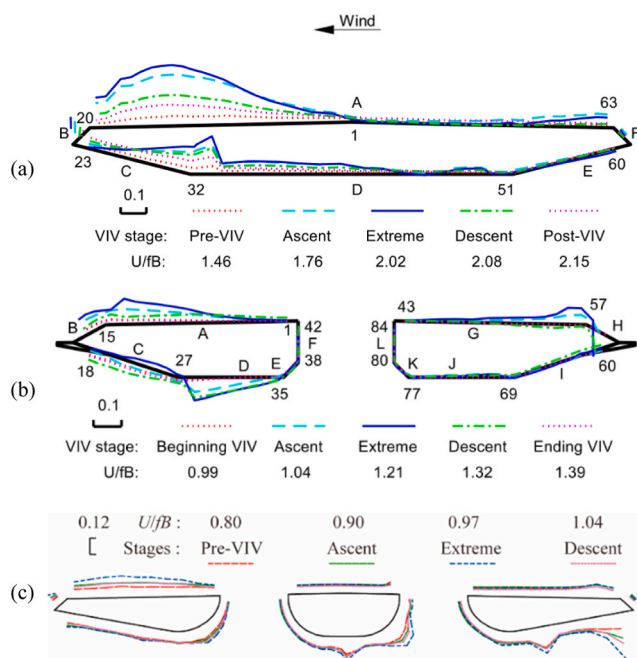


Fig. 18. VIV contributions of distributed aerodynamic forces of the (a) streamlined close-box girder [37], (b) separated twin-box girder [68], (c) split triple-box girder [113].

highway box girder, and rear region were the main causes of torsional VIV.

In summary, the studies mentioned above indicate a close relationship between the VIV of SBGs and the interaction between the gap vortex and the motion of the box girder. The FSI between the unsteady

aerodynamic force, bridge deck vibration, and flow structure during VIV remains unclear. Moreover, it is important to note that the existing studies have been carried out at relatively low Reynolds numbers, which are significantly smaller (2–3 orders of magnitude) than the Reynolds number at full scale. The influence of Reynolds number on the flow mechanism of VIV has yet to be explored. Additionally, the existing studies have predominantly concentrated on specific girder sections, which limits their applicability. Therefore, conducting a comprehensive investigation of base sections is imperative to overcome this limitation. Furthermore, the VIV of SBGs commonly displays multiple lock-in regions, making it essential to understand the distinctions in the excitation mechanisms within these lock-in regions. It is also worth noting that the VIV driving mechanism of the separated triple-box girder is more complex than that of the twin-box one.

5. Control measures for vortex-induced vibration

5.1. Aerodynamic control measures

To ensure the safety, reliability, and comfort of the bridges, various countries' codes have set an amplitude of VIV that is permitted according to the relevant design specifications or standards. If the amplitude exceeds this limit, appropriate vibration control measures such as aerodynamic, mechanical and structural countermeasures should be implemented. Among these measures, aerodynamic control measures are commonly employed to mitigate the VIV of bridges due to their advantages of easy installation, low maintenance requirements, and high effectiveness.

Current studies of VIV aerodynamic control measures for SBGs have primarily focused on passive control. Table 7 summarizes studies on passive aerodynamic control measures for SBGs, such as guide vanes (GVs) [55,116], central grid plates (CGPs) [71,126], vertical central stabilizers (VCSs) [128], vortex generators (VGs) [44], flaps [74] and so on. Some of these measures have been applied to long-span bridges. For

Table 7

Summary of passive aerodynamic control measures for SGBs.

Control methods and sketch	Reference	VIV suppression effect	VIV suppression mechanism	Notes
	Larose et al.[53]	GVs effectively suppress VIV in the section model of Stonecutters Bridge.	GVs reduce the pulsation pressure on the surface of the downstream box girder by injecting high-velocity airflow at the upstream box girder diagonal web to inhibit the formation of vortices. Still, only when the Reynolds number is high enough and the boundary layer thin enough for the GV to be effective.	The Reynolds number of the test should be high enough to ensure adequate flow through the GV.
	Larsen et al.[55]	GVs are ineffective in suppressing vibration at low Reynolds numbers and successfully suppress vertical VIV at high Reynolds numbers.		
	Hui and Yau[41]	GVs successfully suppress the vertical VIV amplitude substantially at a high Reynolds number.		
	Yang et al.[116]	The damper control effect of GV is proportional to the gap ratio, and neither the inner nor outer GV can suppress the VIV of the separated twin-box girder.		
	Zhou et al.[127]	Movable GV effectively suppress the vertical and torsional VIV of a separated twin-box girder.		
	Laima et al.[50]	The light flat plates (0 % porosity) completely suppress the vertical VIV.	CGPs suppress VIV by breaking up the large-scale vortices in the gap, causing the vortices to shrink in scale and weaken in strength, resulting in weakened fluctuations in the lift force and pitching moment of the SGBs.	The suppression effect on the VIV of CGPs is greatly influenced by parameters such as porosity and arrangement. It is crucial to select and verify the design parameters of CGPs to strike a balance between flutter and VIV performance.
	Yang et al.[116]	CGPs significantly reduce the vertical VIV amplitude of the separated twin-box girder, whereas CGPs become more effective as grid porosity decreases.		
	Ma et al.[71]	The suppression effect on VIV is related to the design parameters of CGPs, and an optimal range of porosity and flat plate width ratio exists, as well as a better suppression effect when uniformly distributed and mounted on the upper side.		
	Li et al.[67]	CGPs significantly suppress the VIV of the separated twin-box girder.		
	Zhou et al.[126]	Large porosity is effective in improving flutter and static stability, while smaller porosity is more effective in suppressing torsional VIV.		
	Yang et al.[112]	CGPs with 30 % porosity completely suppress vertical VIV but are ineffective in suppressing torsional VIV at $AOA = +3^\circ$.		
	Duan et al.[24]	As porosity = 25 %~75 %, UCGPs with the non-uniform arrangement are more effective than the uniform arrangement in suppressing vertical VIV.		
	Wu et al.[105]	CGPs effectively suppress torsional VIV and improve the aerodynamic characteristics of separated triple-box girders.		
	He et al.[36]	Decreased porosity notably enhanced the performance under vertical VIV, while the opening size affected the extent of the lock-in region for torsional vibrations.		
	Zhou et al.[128]	The vertical VIV performance improves and then deteriorates as the height of VCSs increases. 0.2 h/H (UVCs) and 0.4 h/H(UVCs) have the best positive effects in suppressing vertical VIV amplitudes. UVCs increase the torsional VIV amplitude, while DVCs suppress the torsional VIV amplitude.		
			VCSs disrupt the motion of vortices on the gap, and the DVCs effectively prevent the interaction of the large-scale vortices on the lower side, thereby weakening the excitation for torsional VIV.	0.2 h/H (DVCs) can meet the requirements of stationary aerodynamic performance, flutter performance and VIV performance for the twin-box girder.
	Kargarmoakhar et al.[44]	VGs can suppress VIV, but their vibration suppression performance tends to be more effective at lower Reynolds numbers.	VGs can effectively suppress vortex shedding and reduce the peak lift spectrum at the lower Reynolds number.	–
	Ma et al.[73]	L-shape flaps can suppress vertical and torsional VIV amplitude under uniform and turbulent flow except for vertical VIV at $AOA = +3^\circ$ under uniform flow.	The interaction between the shedding vortices on the upper and lower surfaces of the upstream box girder is weakened by flaps. They significantly reduce the motion-induced vortex in the gap with the	–
	Duan et al.[23]	The presence of flaps at the lower section of the opening slot is the		

(continued on next page)

Table 7 (continued)

Control methods and sketch	Reference	VIV suppression effect	VIV suppression mechanism	Notes
		primary factor for optimizing the VIV performance.	help of incoming turbulence, effectively suppressing VIV.	

example, guide vanes have been installed on the Stonecutters Bridge [55], and collapsible wind barriers have been implemented on the Xihoumen Bridge [60]. Recently, Zhou et al. [127] designed a series of movable guide vanes, which effectively suppress the vertical and torsional VIV of a separated twin-box girder.

Significantly, the effectiveness of aerodynamic control measures is notably influenced by the Reynolds number. As depicted in Fig. 19(a), Larsen et al. [55] found that the VIV performance of the Stonecutters Bridge deck section deteriorated at a low Reynolds number $Re = 0.33 \times 10^4$ with the presence of the guide vane. However, the vortex vibration was effectively suppressed at a higher Reynolds number $Re = 6.70 \times 10^4$. The boundary layer was measured upstream of the up-wind guide vane for the three Reynolds numbers (Fig. 19). The results highlighted the importance of ensuring sufficiently high test Reynolds numbers to reduce the boundary layer thickness, allowing enough air to pass through the GV's and enhancing their effectiveness. Conversely, some aerodynamic measures exhibit better VIV suppression at low Reynolds numbers than at high Reynolds numbers. Kargarmokhar et al. [44] found that the vortex generators effectively suppressed the VIV of the separated twin-box girder, which was better at a low Reynolds number (9.82×10^4) than at a high Reynolds number (4.74×10^5).

With bridge spans exceeding 2000 m, the control of VIV on bridge decks becomes increasingly challenging. Traditional passive control measures may be inadequate to address the VIV control requirements effectively for the super-long-span bridges. Therefore, there is a need to explore new and efficient aerodynamic control measures. Controlling vortex shedding using spanwise-varying passive or active actuation (three-dimensional control) is a very efficient method for regulating two-dimensional bluff-body wakes [42]. Recent studies have applied three-dimensional control measures to suppress VIV for single-box girders [4,120]. These measures have successfully suppressed the VIV single box girders, providing valuable insights for suppressing VIV for SBGs.

Furthermore, the traditional aerodynamic optimization methods, including wind tunnel tests and numerical simulations, provide numerous data for ML. ML helps aerodynamic optimization to be more efficient, which offers significant advantages in accomplishing such tasks. For long-span bridges with short gap twin-box decks, Cid Montoya et al. [16] employed a surrogate-based strategy to obtain an effective tool for achieving sustainable and safe bridge design by minimizing the

material required to construct the bridge while meeting the performance and safety requirements in the structural, flutter and buffeting design constraints. The strategy consists of two steps: constructing and validating a surrogate model for the bridge deck aerodynamics and the surrogate-based aero-structural optimization of the bridge. The aero-structural optimization framework is shown in Fig. 20. The weakness is that the above aerodynamic optimization framework requires an artificial design series of aerodynamic shapes. Further, combining the optimization framework with ML approaches, especially reinforcement learning, can automatically optimize the aerodynamic shapes to improve optimization efficiency.

To summarize, the effectiveness of passive control measures in suppressing vibrations is notably influenced by the aerodynamic design of bridge decks and the Reynolds number, and the underlying mechanism of vibration suppression through these measures requires further investigation. Developing innovative and efficient aerodynamic control measures for suppressing VIV of SBGs in super-long-span bridge decks is essential. Furthermore, leveraging ML techniques such as deep learning and reinforcement learning can be beneficial in proposing more efficient aerodynamic optimization frameworks.

5.2. Mechanical damping measures

Mechanical damping is an effective measure if aerodynamic control measures fail to meet the performance requirements for VIV. The tuned mass damper (TMD) is a commonly used mechanical damping device in engineering [102,28,95]. However, the application of the TMD in SBGs is challenging. On the one hand, multiple TMDs are needed to control the multi-mode vertical VIV, resulting in an enormous secondary mass requirement for the TMD system. On the other hand, large-span bridges exhibit relatively low vibration frequencies, requiring large static deformation of the TMD springs to suppress low-mode VIVs [9,108].

To address the issue of excessive static deformation in traditional TMDs, Gu et al. [35] proposed the lever-type TMD. Another approach is added as an inerter to the TMD between the secondary mass and the bridge deck, namely the tuned mass damper inerter (TMDI) [34]. Xu et al. [109] and Xu et al. [108] used theoretical analysis and numerical simulation to find that TMDI controls the VIR of long-span bridges with static deformation of springs much smaller than that of the traditional TMD. The efficiency of TMDI depends on the inerter location, mode

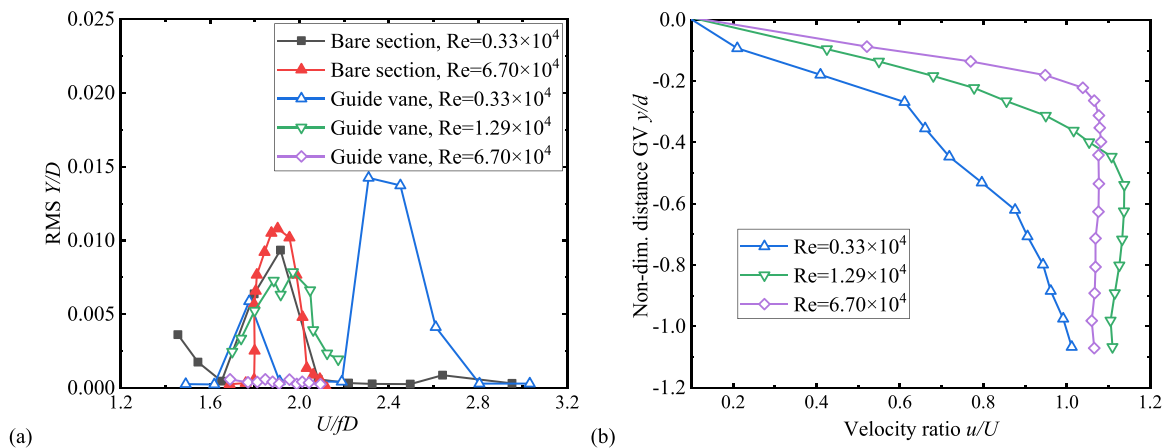


Fig. 19. (a) Vortex response of spring-suspended section models; (b) Nondimensional boundary layer profiles u/U vs. y/d obtained immediately upstream of guide vane [55].

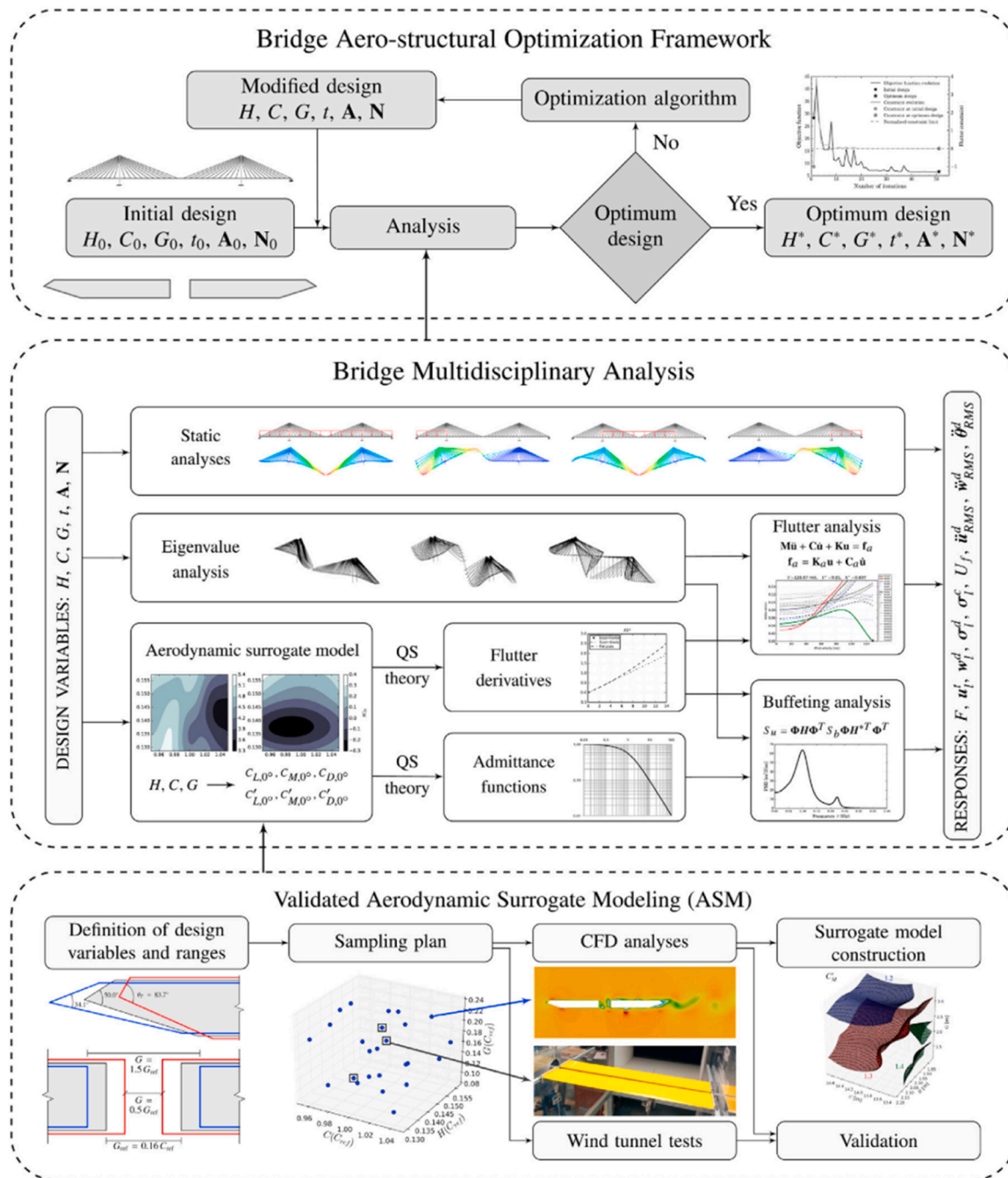


Fig. 20. Flowchart of the aero-structural optimization framework for long-span bridges with short gap twin-box decks considering structural, flutter and buffeting design constraints [16].

shape, physical mass, and inertial mass [17]. Li et al. [66] enhanced the efficiency of vibration control in long-span bridges by combining the lever-arm mechanism with TMDI.

The tower-girder-connected viscous damper has been established as an effective technique for mitigating multi-mode vertical VIV in long-span suspension bridges, as evidenced by its application in the Yingwuzhou Yangtze River Suspension Bridge [10]. Cao et al. [10] determined that viscous dampers positioned near the girder support can effectively increase the modal damping of long-span suspension bridges and suppress vertical VIV, provided there is sufficient distance from the support to the damper attachment point. Furthermore, Cao et al. [9] developed an analytical model to calculate the damping ratio of vertical bending modes, validating the effectiveness of tower-girder-connected dampers in enhancing the modal damping ratio of suspension bridges

and reducing multi-mode vertical VIV. In addition, Chen et al. [12] proposes the damped outriggers for controlling multi-mode vertical VIV. Damped outriggers function by converting the rotational vibrations of the bridge girder into translational displacements at the outrigger tip, utilizing horizontal dampers to dissipate energy and effectively supplement damping, thus mitigating multi-mode vibrations. Based on the Xihoumen bridge, Liu et al. [69] used an optimization algorithm to optimize the parameters of the damped outrigger.

In summary, novel mechanical damping measures, such as TMDI and tower-girder-connected viscous damper, offer promising solutions for the suppression of vertical VIV in long-span bridges, particularly for multi-mode VIVs. Nonetheless, further verification of the efficacy of these novel mechanical damping measures in practical engineering applications is essential. Moreover, considering that the VIV of the SBG is

more severe than that of the single box girder, effectively managing the VIV with dampers remains a considerable challenge.

6. Conclusions and prospects

Due to the aerodynamic interference between the upstream and downstream box girders, SBGs are susceptible to VIV, which has emerged as a major obstacle in the development of super-long-span cable-supported bridges. Consequently, this paper provides a comprehensive overview of the research on the VIV of SBGs, including research methods, influential parameters, excitation mechanisms, and aerodynamic control measures. Despite the extensive research conducted, there are still several areas that require further investigation.

The VIV of SBGs requires the development of various research methods. Firstly, there is a demand for more field monitoring data, coupled with the application of advanced equipment, to accurately determine the actual triggering conditions of VIV and their flow field characteristics. Secondly, wind tunnel tests employing new technologies, such as PIV and active control wind tunnels, could provide valuable insights into the VIV mechanism of the SBGs. Thirdly, FSI simulation using a relatively high-precision CFD simulation, such as LES, helps to reveal the VIV mechanism of the SBGs in detail. Furthermore, developing VIV models that consider nonlinear characteristics and establishing a more accurate method for analyzing three-dimensional VIV are necessary. Finally, ML, widely used in fluid dynamics, presents a promising tool for VIV research. ROM and deep learning methods provide innovative approaches to VIV theory. The deep learning and reinforcement learning frameworks assist in efficiently selecting aerodynamic shapes. Reinforcement learning also holds excellent prospects for the active control of VIV.

A comprehensive exploration of various influence parameters and their mechanisms on VIV will contribute significantly to the academic understanding and practical application of SBGs design and operation. The impact of aerodynamic shape and Reynolds number on the VIV of the SBGs needs further clarification. The characteristics of VIV, the nonlinear VIV, and the VIV mechanism under large AOAs are unclear. The suppression mechanism of turbulence on the VIV of the SBGs also needs further clarification. In particular, the effect of the turbulence integral length scale on the VIV remains unclear. Finally, future studies need to optimize the geometric details and the placement of attachments to minimize their adverse effects on VIVs.

Understanding the VIV excitation mechanisms of the SBGs from a flow field perspective is crucial for optimizing the aerodynamic shapes of bridge decks and proposing effective aerodynamic control measures. The VIV mechanism, particularly for the triple-box girders, requires further clarification. Current VIV studies are predominantly based on specific bridge sections, underscoring the need for systematic research on foundational sections. Given that SBGs often exhibit multiple lock-in regions, it is crucial to elucidate the differences in the VIV mechanism across these regions.

Further investigation of the aerodynamic control measures for VIV of SBGs is also required. The effectiveness of these measures is closely related to the shapes of the bridge decks and the Reynolds number. Studying the VIV suppression mechanism of passive control measures provides a theoretical foundation for selecting parameters of aerodynamic control measures. Additionally, there is a need to explore novel aerodynamic control measures, such as three-dimensional control, active control, and semi-active aerodynamic control measures. Take advantage of ML methods such as deep learning and reinforcement learning to propose more efficient aerodynamic optimization frameworks.

In addition, it is worth noting that most current studies on SBGs tend to focus on VIV and flutter performance separately. However, conflicts between the two can arise, as demonstrated by the effect of the porosity of the central grid plates. Therefore, a comprehensive evaluation of both types of aeroelasticity problems is necessary to fully understand the

behavior of SBGs under different conditions. Furthermore, it is crucial to consider VIV performance during the construction stage of long-span bridges. The construction process can take a long time, and various parameters, such as structural stiffness, damping, and aerodynamic shapes, can differ significantly from when the bridge is complete. Therefore, it is essential to factor in these variables to ensure the bridge can withstand VIV during construction and after completion.

CRediT authorship contribution statement

Haotian Dong: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Xiaoqing Du:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization. **Hongyu Zhu:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] AIJ, Guidebook of recommendation for loads on buildings 2: Wind-induced response and load estimation/Practical guides of CFD for wind resistant design (AIJ Tokyo, 2017).
- [2] Álvarez AJ, Nieto F, Kwok KCS, Hernández S. Aerodynamic performance of twin-box decks: a parametric study on gap width effects based on validated 2D URANS simulations. *J Wind Eng Ind Aerodyn* 2018;182:202–21.
- [3] Álvarez AJ, Nieto F, Kwok KCS, Hernández S. A computational study on the aerodynamics of a twin-box bridge with a focus on the spanwise features. *J Wind Eng Ind Aerodyn* 2021;209:104465.
- [4] Bai H, Ji NC, Xu GJ, Li JW. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: Sealed traffic barrier. *J Wind Eng Ind Aerodyn* 2020;206:104302.
- [5] Belloli M, Fossati F, Giappino S, Muggiasca S. Vortex induced vibrations of a bridge deck: Dynamic response and surface pressure distribution. *J Wind Eng Ind Aerodyn* 2014;133:160–8.
- [6] Brancaloni F, Diana G. The aerodynamic design of the messina straits bridge. *J Wind Eng Ind Aerodyn* 1993;48:395–409.
- [7] Bruno L, Salvetti MV, Ricciardelli F. Benchmark on the aerodynamics of a rectangular 5:1 cylinder: an overview after the first four years of activity. *J Wind Eng Ind Aerodyn* 2014;126:87–106.
- [8] Cao F, Ge Y. Air-induced nonlinear damping and added mass of vertically vibrating bridge deck section models under zero wind speed. *J Wind Eng Ind Aerodyn* 2017;169:217–31.
- [9] Cao Y, Huang Z, Hua X, Chen Z, Zhang H, Wang W. Analytical model analysis of suspension bridges with tower-girder-connected dampers. *Eng Struct* 2025;322.
- [10] Cao Y, Huang Z, Zhang H, Hua X, Chen Z, Wan T. Discrete viscous dampers for multi-mode vortex-induced vibration control of long-span suspension bridges. *J Wind Eng Ind Aerodyn* 2023;243.
- [11] Chen, A., Ma, R., Wang, D., Liu, G., Ai, H., Liu, T., 2018. Wind-resistant design specification for highway bridges. Ministry of Communications of the People's Republic of China.
- [12] Chen L, Liu Z, Nagarajaiah S, Sun L, Zhao L, Cui W. Vibration mitigation of long-span bridges with damped outriggers. *Eng Struct* 2022;271.
- [13] Chen W, Li H, Hu H. An experimental study on the unsteady vortices and turbulent flow structures around twin-box-girder bridge deck models with different gap ratios. *J Wind Eng Ind Aerodyn* 2014;132:27–36.
- [14] Chen X, Ma C, Xian R, Liu T. Effects of gap configuration on vortex-induced vibration of twin-box girder and countermeasure study. *J Wind Eng Ind Aerodyn* 2022;231:105232.
- [15] Chen X, Zhu C, Ma C, Li Q. Vortex-induced vibration of twin-box girder and its aerodynamic characteristics study based on high-order dynamic decomposition. *J Wind Eng Ind Aerodyn* 2024;249.
- [16] Cid Montoya M, Nieto F, Hernández S, Fontán A, Jurado JA, Kareem A. Optimization of bridges with short gap streamlined twin-box decks considering structural, flutter and buffeting performance. *J Wind Eng Ind Aerodyn* 2021;208:104316.

- [17] Dai J, Xu Z-D, Gai P-P, Hu Z-W. Optimal design of tuned mass damper inerter with a Maxwell element for mitigating the vortex-induced vibration in bridges. *MECH SYST SIGNAL PR* 2021;148.
- [18] Deng X, Hu G, Chen W. Intelligent active flow control of long-span bridge deck using deep reinforcement learning integrated transfer learning. *J Wind Eng Ind Aerodyn* 2024;244.
- [19] Diana G, Fiammenghi G, Belloli M, Rocchi D. Wind tunnel tests and numerical approach for long span bridges: The Messina bridge. *J Wind Eng Ind Aerodyn* 2013;122:38–49.
- [20] Diana G, Resta F, Belloli M, Rocchi D. On the vortex shedding forcing on suspension bridge deck. *J Wind Eng Ind Aerodyn* 2006;94:341–63.
- [21] Dong H, Shi D, Du X. Flow around single and two tandem rectangular cylinders with various single-side fairings. *Phys Fluids* 2024;36.
- [22] Dong L, Jia J, Zhang L. Numerical analysis on spanwise correlation of vortex-induced force of split double-box beam. *Appl Sci* 2022;12:12688.
- [23] Duan Q-S, Shang J-m, Ma C-m, Ding Z, Pei C. Influence of flow restrictor plate on vertical vortex-induced vibration performance of twin-box girder. *Structures* 2024;67.
- [24] Duan Q, Ma C, Yang H. Influence of non-uniform grid plates on the vortex-induced vibration performance of twin-box girder section. *Structures* 2023;58.
- [25] Ehsan F, Scanlan RH. Vortex-induced vibrations of flexible bridges. *J Eng Mech* 1990;116:1392–411.
- [26] Franke, J., Hirsch, C., Jensen, A., Krüsi, H., Schatzmann, M., Westbury, P., Miles, S., Wisse, J., Wright, N., *Recommendations on the use of CFD in predicting pedestrian wind environment* (2004).
- [27] Fujino Y, Siringoringo D. Vibration mechanisms and controls of long-span bridges: a review. *Struct Eng Int* 2013;23:248–68.
- [28] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay Crossing bridge. *J Struct Eng* 2002;128:1012–25.
- [29] Gao D, Deng Z, Yang W, Chen W. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: A vortex-dynamics approach. *J Fluids Struct* 2021;105:103348.
- [30] Ge B, Ma R, Li F, Hu X, Chen A. Probabilistic vortex-induced vibration occurrence prediction of the twin-box girder for long-span cable-stayed bridges based on wind tunnel tests. *Eng Struct* 2022;262:114325.
- [31] Ge, Y., Shao, Y., *3D nonlinear aerostatic stability analysis for suspension bridges with ultra-long span* (2013).
- [32] Ge, Y., Yang, Y., Cao, F., *VIV sectional model testing and field measurement of Xihoumen Suspension Bridge with twin box girder* (2011).
- [33] Ge Y, Zhao L, Xu K. Review and reflection on vortex-induced vibration of main girders of long-span bridges. *China J Highw* 2019;32:1–18.
- [34] Giaralis A, Petrini F. Wind-induced vibration mitigation in tall buildings using the tuned mass-damper-inerter. *J Struct Eng* 2017;143.
- [35] Gu M, Chen S, Chang C. Control of wind-induced vibrations of long-span bridges by semi-active lever-type TMD. *J Wind Eng Ind Aerodyn* 2002;90:111–26.
- [36] He H, Wang J, Wang F. Investigation of the effects of vortex isolation plates with different opening ratios and sizes on vortex-induced vibration. *Appl Sci* 2024;14.
- [37] Hu C, Zhao L, Ge Y. Time-frequency evolutionary characteristics of aerodynamic forces around a streamlined closed-box girder during vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2018;182:330–43.
- [38] Hu C, Zhao L, Ge Y. Multiple-order vertical vortex-induced vibration mechanism of a typical streamlined closed-box girder. *J Wind Eng Ind Aerodyn* 2022;227.
- [39] Hu C, Zhao L, Ge YJW. Structures. A Simpl Vortex Model Mech Vortex-Induc Vib A Streamline closed-Box girder 2021;32:309–19.
- [40] Huang Z, Li Y, Hua X, Chen Z, Wen Q. Automatic identification of bridge vortex-induced vibration using random decrement method. *Appl Sci* 2019;9:2049.
- [41] Hui MC, Yau DM. The aerodynamic behaviour of the deck of stonecutters Bridge, Hong Kong. *Struct Eng Int* 2010;20:79–90.
- [42] Hwang Y, Kim J, Choi H. Stabilization of absolute instability in spanwise wavy two-dimensional wakes. *J Fluid Mech* 2013;727:346–78.
- [43] Hwang YC, Kim S, Kim HK. Cause investigation of high-mode vortex-induced vibration in a long-span suspension bridge. *Struct Infrastruct Eng* 2019;16:84–93.
- [44] Kargarmokhar R, Chowdhury AG, Irwin PA. Reynolds number effects on twin box girder long span bridge aerodynamics. *WIND STRUCT* 2015;20:327–47.
- [45] Kim S, Kim HK, Hwang YC. Enhanced damping estimation for cable-stayed bridges based on operational monitoring data. *STRUCT ENG INT* 2018;28:308–17.
- [46] Kroon IB, Polk H, Fuglsang K. 1915 Çanakkale Bridge—Meeting the Challenge. Springer; 2021.
- [47] Kwok KCS, Qin XR, Fok CH, Hitchcock PA. Wind-induced pressures around a sectional twin-deck bridge model: Effects of gap-width on the aerodynamic forces and vortex shedding mechanisms. *J Wind Eng Ind Aerodyn* 2012;110:50–61.
- [48] Laima S, Jiang C, Li H, Chen W, Ou J. A numerical investigation of Reynolds number sensitivity of flow characteristics around a twin-box girder. *J Wind Eng Ind Aerodyn* 2018;172:298–316.
- [49] Laima S, Li H. Effects of gap width on flow motions around twin-box girders and vortex-induced vibrations. *J Wind Eng Ind Aerodyn* 2015;139:37–49.
- [50] Laima S, Li H, Chen W, Li F. Investigation and control of vortex-induced vibration of twin box girders. *J Fluids Struct* 2013;39:205–21.
- [51] Laima S, Li H, Chen W, Ou J. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J Fluids Struct* 2018;77:115–33.
- [52] Laima S, Wu B, Jiang C, Chen W, Li H. Numerical study on Reynolds number effects on the aerodynamic characteristics of a twin-box girder. *WIND STRUCT* 2019;28:285–98.
- [53] Larose, G., Larsen, S., Larsen, A., Hui, M., Jensen, A., *Sectional model experiments at high Reynolds number for the deck of a 1018 m span cable-stayed bridge* (Lubbock TX, USA, 2003).
- [54] Larose GL, D'Auteuil A. On the Reynolds number sensitivity of the aerodynamics of bluff bodies with sharp edges. *J Wind Eng Ind Aerodyn* 2006;94:365–76.
- [55] Larsen A, Savage M, Lafrenière A, Hui MCH, Larsen SV. Investigation of vortex response of a twin box bridge section at high and low Reynolds numbers. *J Wind Eng Ind Aerodyn* 2008;96:934–44.
- [56] Larsen A, Walther JH. Aeroelastic analysis of bridge girder sections based on discrete vortex simulations. *J Wind Eng Ind Aerodyn* 1997;67-68:253–65.
- [57] Lee S, Kwon SD, Yoon J. Reynolds number sensitivity to aerodynamic forces of twin box bridge girder. *J Wind Eng Ind Aerodyn* 2014;127:59–68.
- [58] Li H, Laima S, Jing H. Reynolds number effects on aerodynamic characteristics and vortex-induced vibration of a twin-box girder. *J Fluids Struct* 2014;50:358–75.
- [59] Li H, Laima S, Ou J, Zhao X, Zhou W, Yu Y, Li N, Liu Z. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Eng Struct* 2011;33:1894–907.
- [60] Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2014;124:54–67.
- [61] Li H, Ou J. The state of the art in structural health monitoring of cable-stayed bridges. *J Civ Struct Health* 2015;6:43–67.
- [62] Li S, Laima S, Li H. Cluster analysis of winds and wind-induced vibrations on a long-span bridge based on long-term field monitoring data. *Eng Struct* 2017;138:245–59.
- [63] Li S, Laima S, Li H. Data-driven modeling of vortex-induced vibration of a long-span suspension bridge using decision tree learning and support vector regression. *J Wind Eng Ind Aerodyn* 2018;172:196–211.
- [64] Li S, Laima S, Li H. Physics-guided deep learning framework for predictive modeling of bridge vortex-induced vibrations from field monitoring. *Phys Fluids* 2021;33.
- [65] Li S, Li X, Li J, Su Y, Wu B. Measurements of the unsteady lift force on a 5: 1 rectangular cylinder based on active-passive hybrid wind tunnel tests. *MEASUREMENT* 2023;213:112674.
- [66] Li Z, Xu K, Ma R, Bi K, Han Q. A novel lever-arm tuned mass damper inerter (LTMDI) for vibration control of long-span bridges. *Eng Struct* 2023;293.
- [67] Li ZG, Zhou Q, Liao HL, Ma CM. Numerical studies of the suppression of vortex-induced vibrations of twin box girders by central grids. *WIND STRUCT* 2018;26:305–15.
- [68] Liu S, Zhao L, Fang G, Hu C, Ge Y. Investigation on aerodynamic force nonlinear evolution for a central-slotted box girder under torsional vortex-induced vibration. *J Fluids Struct* 2021;106:103380.
- [69] Liu Z, Chen L, Sun L, Zhao L, Cui W, Guan H. Multimode damping optimization of a long-span suspension bridge with damped outriggers for suppressing vortex-induced vibrations. *Eng Struct* 2023;286.
- [70] Ma C, Pei C, Liao H, Liu M, Li M. Field measurement and wind tunnel study of aerodynamic characteristics of twin-box girder. *J Wind Eng Ind Aerodyn* 2020;202:104209.
- [71] Ma CM, Wang JX, Li QS, Qin H, Liao HL. Vortex-induced vibration performance and suppression mechanism for a long suspension bridge with wide twin-box girder. *J Struct Eng* 2018;144:04018202.
- [72] Ma K, Hu C, Zhou Z. Investigation on vortex-induced vibration of twin rectangular 5:1 cylinders through wind tunnel tests and POD analysis. *J Wind Eng Ind Aerodyn* 2019;187:97–107.
- [73] Ma R, Li F, Hu X, Ge B, He Q. Experiment study on vortex-induced vibration of a twin-box girder with large cantilever plates. *Structures* 2023;55:891–907.
- [74] Ma R, Li F, Hu X, Ge B, He Q. Experiment study on vortex-induced vibration of a twin-box girder with large cantilever plates. Elsevier; 2023.
- [75] Marra AM, Mannini C, Bartoli G. Van der Pol-type equation for modeling vortex-induced oscillations of bridge decks. *J Wind Eng Ind Aerodyn* 2011;99:776–85.
- [76] Matsuda K, Cooper K, Tanaka H, Tokushige M, Iwasaki T. An investigation of Reynolds number effects on the steady and unsteady aerodynamic forces on a 1: 10 scale bridge deck section model. *J Wind Eng Ind Aerodyn* 2001;89:619–32.
- [77] Meng H, Chen G, Gao D. Aerodynamics and surrounding flow patterns of a long-span bridge girder model with triple-separated boxes. *Phys Fluids* 2024;36.
- [78] Meng H, Chen W, Duan Y, Li H, Gao D. Gap effects on the aerodynamic characteristics around three rectangular boxes in tandem arrangement. *Phys Fluids* 2022;34:105124.
- [79] Meng H, Chen W, Li H, Gao D. The vortex-induced vibrations of a triple-box girder model of a long-span rail-cum-road bridge. *J Wind Eng Ind Aerodyn* 2023;235:105359.
- [80] Meng H, Qiu C, Yang W, Gao D. Complex flow dynamics for a static triple-box girder under various angles of attack. *Phys Fluids* 2024;36.
- [81] Miranda Sd, Patruno L, Ricci M, Ubertini F. Numerical study of a twin box bridge deck with increasing gap ratio by using RANS and LES approaches. *Eng Struct* 2015;99:546–58.
- [82] Montoya MC, Bai H, Ye M. Shaping bridge decks for VIV mitigation: A wind tunnel data-driven adaptive surrogate-based optimization method. *J Wind Eng Ind Aerodyn* 2023;242.
- [83] Noguchi K, Ito Y, Yagi T. Numerical evaluation of vortex-induced vibration amplitude of a box girder bridge using forced oscillation method. *J Wind Eng Ind Aerodyn* 2020;196.
- [84] Ran Y, Chen W-L, Jin Y, Li H, Gao D. Reynolds number sensitivity of the aerostatic characteristics of a long-span rail-cum-road bridge girder with three separated boxes. *J Wind Eng Ind Aerodyn* 2024;250.

- [85] Ran Y, Chen W-L, Jin Y, Li H, Gao D. Reynolds number sensitivity of the vortex dynamics around a long-span rail-cum-road bridge girder with three separated boxes. *J Wind Eng Ind Aerodyn* 2024;250.
- [86] Sarpkaya T. A critical review of the intrinsic nature of vortex-induced vibrations. *J Fluids Struct* 2004;19:389–447.
- [87] Sarwar MW, Ishihara T. Numerical study on suppression of vortex-induced vibrations of box girder bridge section by aerodynamic countermeasures. *J Wind Eng Ind Aerodyn* 2010;98:701–11.
- [88] Sato H, Hirahara N, Fumoto K, Hirano S, Kusahara S. Full aeroelastic model test of a super long-span bridge with slotted box girder. *J Wind Eng Ind Aerodyn* 2002;90:2023–32.
- [89] Scanlan, R.H., State-of-the-art methods for calculating flutter, vortex-induced, and buffeting response of bridge structures Document No. Number, 1981.
- [90] Schewe G, Larsen A. Reynolds number effects in the flow around a bluff bridge deck cross section. *J Wind Eng Ind Aerodyn* 1998;74-6:829–38.
- [91] Scruton, C., *On the wind excited oscillations of stacks, towers and masts* (1963).
- [92] Shang JM, Zhou Q, Liao HL, Larsen A, Wang J, Li MS. Numerical study of wake and aerodynamic forces on a twin-box bridge deck with different gap ratios. *WIND STRUCT* 2020;30:367–78.
- [93] Song Y, Ti Z, Li Y. An efficient two-stage hybrid framework to evaluate vortex-induced vibration for bridge deck based on divergent vibration. *J Wind Eng Ind Aerodyn* 2023;233.
- [94] Sun H, Zhu L, Zhu Q, Qian C, Meng X, Du L. Identification of parameters for same-order nonlinear damping terms in polynomial-type vortex-induced force models for bridge decks. *J Wind Eng Ind Aerodyn* 2022;229:105158.
- [95] Sun Z, Zou Z, Ying X, Li X. Tuned mass dampers for wind-induced vibration control of chongqi bridge. *J Bridge Eng* 2020;25.
- [96] Vandiver JK. Damping Parameters for flow-induced vibration. *J Fluids Struct* 2012;35:105–19.
- [97] Vickery B, Watkins R. Flow-induced vibrations of cylindrical structures. *Hydraulics and Fluid Mechanics*. Elsevier; 1964. p. 213–41.
- [98] Wang C, Hua X, Feng Z, Xiao H, Chen Z, Zhang J. Experimental investigation on vortex-induced vibrations of a triple-box girder with web modification. *J Wind Eng Ind Aerodyn* 2021;218:104783.
- [99] Wang C, Huang Z, Hua X, He X, Zhou S, Chen Z. Aerodynamic mechanism of triggering and suppression of vortex-induced vibrations for a triple-box girder. *J Wind Eng Ind Aerodyn* 2022;227:105051.
- [100] Wang Y, Chen X. Extraction of aerodynamic damping and prediction of vortex-induced vibration of bridge deck using CFD simulation of forced vibration. *J Wind Eng Ind Aerodyn* 2022;224.
- [101] Watanabe S, Fumoto K. Aerodynamic study of slotted box girder using computational fluid dynamics. *J Wind Eng Ind Aerodyn* 2008;96:1885–94.
- [102] Weber F, Mašlanka M. Frequency and damping adaptation of a TMD with controlled MR damper. *SMART MATER STRUCT* 2012;21.
- [103] Wu B, Laima S. Experimental study on characteristics of vortex-induced vibration of a twin-box girder and damping effects. *J Fluids Struct* 2021;103:103282.
- [104] Wu B, Zhang L, Yang Y, Liu L, Ni Z. Refined time-domain buffeting analysis of a long-span suspension bridge in mountainous urban terrain. *Adv Civ Eng* 2020; 2020:4703169.
- [105] Wu F, Wang Z, Zhao L, Pan T, Xiao H, Ge Y. Aerodynamic force distribution characteristics around a double-slotted box girder of a long-span bridge during vortex-induced vibration. *J Bridge Eng* 2023;28:04022132.
- [106] Wu T, Kareem A. An overview of vortex-induced vibration (VIV) of bridge decks. *Front Struct Civ Eng* 2012;6:335–47.
- [107] Wu T, Kareem A. Vortex-induced vibration of bridge decks: volterra series-based model. *J Eng Mech* 2013;139:1831–43.
- [108] Xu K, Bi K, Ge Y, Zhao L, Han Q, Du X. Performance evaluation of inerter-based dampers for vortex-induced vibration control of long-span bridges: A comparative study. *STRUCT CONTROL HLTH* 2020;27.
- [109] Xu K, Bi K, Han Q, Li X, Du X. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: Analytical study. *Eng Struct* 2019;182: 101–11.
- [110] Xu K, Zhao L, Ge Y. Reduced-order modeling and calculation of vortex-induced vibration for large-span bridges. *J Wind Eng Ind Aerodyn* 2017;167:228–41.
- [111] Xu S, Ma R, Wang D, Chen A, Tian H. Prediction analysis of vortex-induced vibration of long-span suspension bridge based on monitoring data. *J Wind Eng Ind Aerodyn* 2019;191:312–24.
- [112] Yang F, Zheng S, Yan Z. Vortex-induced Vibration And Control Of Split Three-box Girder Bridges. *STRUCT ENG INT* 2021;1–10.
- [113] Yang FF, Zheng SX, Zhou Q, Yan ZX, Ding ZH, Tai XY. Vortex-excited force evolutionary characteristics of split three-box girder bridges during vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2021;218:104762.
- [114] Yang Y, Ma T, Ge Y. Evaluation on bridge dynamic properties and VIV performance based on wind tunnel test and field measurement. *WIND STRUCT* 2015;20:719–37.
- [115] Yang Y, Zhou R, Ge Y, Mohotti D, Mendis P. Aerodynamic instability performance of twin box girders for long-span bridges. *J Wind Eng Ind Aerodyn* 2015;145: 196–208.
- [116] Yang Y, Zhou R, Ge Y, Zhang L. Experimental studies on VIV performance and countermeasures for twin-box girder bridges with various slot width ratios. *J Fluids Struct* 2016;66:476–89.
- [117] Yuan W, Laima S, Chen W, Li H, Hu H. Investigation on the vortex-and-wake-induced vibration of a separated-box bridge girder. *J Fluids Struct* 2017;70: 145–61.
- [118] Zasso A, Argentini T, Omarini S, Rocchi D, Øiseth O. Peculiar aerodynamic advantages and problems of twin-box girder decks for long span crossings. *Bridge Struct* 2019;15:111–20.
- [119] Zdravkovich MM. *Flow Around Circular Cylinders Vol 1: Fundamentals*. Oxford University Press; 1997.
- [120] Zhan J, Xin DB, Ou JP, Liu ZW. Experimental study on suppressing vortex -induced vibration of a long -span bridge by installing the wavy railings. *J Wind Eng Ind Aerodyn* 2020;202.
- [121] Zhang J, Wang W, Wang C, Hua X, Huang Z. Numerical investigation on vertical vortex-induced vibrations of triple-box girders with curved and linear-type webs. *J Wind Eng Ind Aerodyn* 2023;242.
- [122] Zhang Y, Cardiff P, Keenahan J. Wind-induced phenomena in long-span cable-supported bridges: a comparative review of wind tunnel tests and computational fluid dynamics modelling. *Appl Sci* 2021;11.
- [123] Zhang Z, Xu F. Added mass and damping effects on vibrating bridge decks in still air. *J Wind Eng Ind Aerodyn* 2019;191:227–38.
- [124] Zhang Z, Zhang X, Yang Y, Ge Y. Nonlinear aerodynamic and energy input properties of a twin-box girder bridge deck section. *J Fluids Struct* 2017;74: 413–26.
- [125] Zhao L, Wu F, Han T, Li L, Pan T, Xiao H, Ge Y. Aerodynamic force distribution and vortex drifting pattern around a double-slotted box girder under vertical vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2023;241.
- [126] Zhou R, Ge Y, Liu Q, Yang Y, Zhang L. Experimental and numerical studies of wind-resistance performance of twin-box girder bridges with various grid plates. *THIN WALL STRUCT* 2021;166:108088.
- [127] Zhou R, Lu P, Gao X, Ge Y, Yang Y, Zhang L. Role of moveable guide vane with various configurations in controlling the vortex-induced vibration of twin-box girder suspension bridges: an experimental investigation. *Eng Struct* 2023;281: 115762.
- [128] Zhou R, Yang Y, Ge Y, Zhang L. Comprehensive evaluation of aerodynamic performance of twin-box girder bridges with vertical stabilizers. *J Wind Eng Ind Aerodyn* 2018;175:317–27.
- [129] Zhu L-D, Xu Y-L, Guo Z, Chang G-Z, Tan X. Yaw wind effect on flutter instability of four typical bridge decks. *WIND STRUCT* 2013;17:317–43.
- [130] Zhu Q, Chen BY, Zhu LD, Xu YL. Investigation on characteristics and span-wise correlation of vortex-induced forces on a twin-box deck using newly-developed wind-tunnel test technique. *J Wind Eng Ind Aerodyn* 2017;164:69–81.
- [131] Zhu Q, Xu YL, Zhu LD, Chen BY. A semi-empirical model for vortex-induced vertical forces on a twin-box deck under turbulent wind flow. *J Fluids Struct* 2017;71:183–98.
- [132] Zhu Q, Xu YL, Zhu LD, Li H. Vortex-induced vibration analysis of long-span bridges with twin-box decks under non-uniformly distributed turbulent winds. *J Wind Eng Ind Aerodyn* 2018;172:31–41.